

REPORT OF

SUBSURFACE EXPLORATION AND
GEOTECHNICAL ENGINEERING ANALYSIS

PROPOSED BUSINESS PARK
ZPG NITTO DENKO REMNANT PROPERTY
SWC 108TH STREET AND HIGHWAY 31
PLEASANT PRAIRIE, WISCONSIN
ECS PROJECT NO. 42:1291

FOR

ZILBER PROPERTY GROUP
KENOSHA, WISCONSIN

AUGUST 25, 2016

ECS MIDWEST, LLC
"Setting the Standard for Service"
Geotechnical • Construction Materials • Environmental • Facilities

August 25, 2016

Mr. Jason Lueders
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cc: Pinnacle Engineering Group
Attention: Mr. Adam Artz, P.E.
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ECS Project No. 42:1267

Reference: Proposal for Subsurface Exploration and Geotechnical Engineering Services
Proposed Business Park
ZPG Nitto Denko Remnant Property
SWC 108th Street and Highway 31
Village of Pleasant Prairie, Wisconsin

Dear Mr. Lueders,

As authorized by your acceptance of ECS' Proposal No. 42:853-GP (dated August 3, 2016) and your Purchase Order No. CM7010-000057, dated 8/5/2016, ECS Midwest, LLC (ECS) has completed the subsurface exploration and geotechnical engineering analyses for the proposed Business Park development to be constructed to the southwest of the intersection of 108th Street and Highway 31 in Pleasant Prairie, Wisconsin.

A report, including the results of our subsurface exploration, boring data, laboratory testing, recommendations regarding the geotechnical engineering and construction aspects for the project, a Site Location Diagram, a Boring Location Diagram and a Test Pit Location Plan are enclosed herein. The recommendations presented herein are intended for use by your office and for use by other professionals involved in the design and construction stages of the project.

We appreciate the opportunity to be of service to Zilber Property Group and Pinnacle Engineering Group on this project. If you have questions with regard to the information and recommendations contained in this report, or if we may be of further service to you during the planning and/or construction phase of this project, please do not hesitate to contact the undersigned.

Respectfully,

ECS MIDWEST, LLC

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Principal Engineer

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REPORT

PROJECT

Subsurface Exploration and
Geotechnical Engineering Analysis
Proposed Business Park
ZPG Nitto Denko Remnant Property
SWC 108th Street and Highway 31
Pleasant, Wisconsin

CLIENT

Zilber Property Group
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SUBMITTED BY

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PROJECT NO. 42:1291

DATE August 25, 2016

TABLE OF CONTENTS

EXECUTIVE SUMMARY

	Page
PROJECT OVERVIEW	**1**
Introduction	1
Site Location	1
Existing Site Conditions and Past History	1
Proposed Construction	2
Purpose of Exploration and Scope of Services	3
EXPLORATION PROCEDURES	**5**
Subsurface Exploration Procedures	5
Laboratory Testing Program	6
EXPLORATION RESULTS	**7**
Soil Conditions	7
Groundwater Observations	8
ANALYSIS AND RECOMMENDATIONS	**9**
Overview	9
Subgrade Preparation and Earthwork Operations	9
Fill Placement	12
Foundation Recommendations	13
Floor Slab Design	14
Underslab Drainage Considerations	15
Below-Grade Loading Dock Walls	15
Pavement Design Recommendations	17
Pavement Maintenance	19
Detention Pond Considerations	19
PROJECT CONSTRUCTION RECOMMENDATIONS	**21**
General Construction Considerations	21
Construction Dewatering	21
Closing	21

APPENDIX

EXECUTIVE SUMMARY

The subsurface conditions encountered during our exploration and ECS' conclusions and recommendations are summarized below. This summary should not be considered apart from the entire text of the report with all the qualifications and considerations mentioned herein. Details of our conclusions and recommendations are discussed in the following sections and in the Appendix of this report.

The project site is located at the southwest corner of the intersection of 108th Street and Highway 31, in the Village of Pleasant Prairie, Kenosha County, Wisconsin. The majority of the site is currently undeveloped and vacant and the southeast portion of the site is used for agricultural purposes (soybeans). Existing wetlands are located along the western and southwestern portions of the site and extend in a northeast/southwest direction through the southeastern portion of the site (following an existing ditch). In addition, an artificial wetland area (consisting of a former sediment basin) is located in the west-central portion of the site and is partially located within one of the proposed building areas.

We understand the proposed construction will consist of two (2) one-story warehouse/manufacturing type buildings (designated as Facilities "I" and "II") and the buildings will have precast tilt-up walls with slabs-on-grade. Each building will have recessed loading docks, truck parking areas and automobile parking lots. Based on verbal information regarding finished first floor elevations (FFE) provided by Pinnacle Engineering Group and existing site grades, we anticipate up to about 4 to 6 feet of cut and 6 to 8 feet of fill will be required to establish the Facility "I" subgrade elevation and about 5 feet of cut and fill will be required to establish the Facility "II" subgrade elevation, exclusive of site preparation requirements. To better understand the subsurface conditions at the project site a series of thirteen (13) soil borings and nine (9) backhoe test pits were performed. The subsurface conditions encountered at the onsite borings and test pits can be summarized as follows.

The surficial materials encountered at the test boring and test pit locations consists of topsoil ranging in thickness from about 6 to 36 inches (typically 8 to 16 inches). Silty CLAY FILL was encountered in the majority of the Facility "I" borings and test pits to depths ranging from about 2 to 4½ feet below existing grades. No discernable fill soils were encountered at the Facility "II" boring or test pit locations. At boring B-2, a possible buried topsoil layer (about 6 inches thick) was encountered below the fill and extended to a depth of about 5 feet. Native soils underlying the surface materials, fill soils and possible buried topsoil consist of predominantly Silty CLAY and Clayey SILT extending to at least the maximum depth explored, 21 feet. Interbedded SILT, Silty SAND, Sandy SILT or Clayey SAND layers were encountered at several borings. Possible cobbles or boulders were encountered at boring B-8 at about 4 to 5 feet.

The existing FILL soils (Silty CLAY and Clayey SILT) typically exhibited an unconfined compressive strength value (Qp value) greater than 4½ tons per square foot (tsf), indicative of a hard consistency for cohesive materials. The surficial FILL soils were relatively dry to moist at the time of the subsurface explorations with moisture contents ranging from about 8 to 18 percent (typically 11 to 13 percent). The natural Silty CLAY and Clayey SILT soils exhibited Qp values in the range of 1½ to greater than 4½ tsf, (stiff to hard consistencies) and moisture contents in the range of 11 to 27 percent (typically 12 to 20 percent). The natural SILT, Silty SAND, Sandy SILT or Clayey SAND exhibited SPT N-values ranging from about 10 to 29 blows per foot (bpf) (medium dense).

Test Pit Nos. TP-2, TP-3, TP-7 and TP-9 were performed within or near the planned stormwater detention pond areas. The surface materials at those test pits consisted of about 10 to 36 inches of topsoil (USDA classification of Silty Clay Loam). Existing fill classified as Sandy CLAY (USDA classification of Sandy Clay) was encountered at Test Pit TP-7 to a depth of about 2½ feet below the ground surface. Native soils below the fill at Test Pit TP-7 and below the surface topsoil at Test Pit Nos. TP-2, TP-3 and TP-9 consist of predominantly Silty CLAY to Clayey SILT (USDA classification of Silty Clay, Silty Clay Loam and Clay Loam) extending to the maximum depths explored at those locations ranging from about 6 to 8½ feet below existing grades.

Based on the results of this exploration, the long-term groundwater level may be located at a depth ranging from approximately 7 to 14½ feet below existing site grades which corresponds to approximately between EL. +698½ to EL. +716 feet.

Based on information provided by Pinnacle Engineering Group we understand that the north and northwest sides of the site were previously mass graded and the grading activities were reportedly supervised by a geotechnical testing agency. However, no information regarding the limits of fill placement throughout other areas of the site or field reports documenting the placement, compaction and testing of on-site existing fill soils was provided at the time of this report preparation. When and if any information becomes available, we request it be provided to ECS for review and possible revision to the conclusions and recommendations included in this report. Based on the results of the borings and test pits, the fill soils exhibit relatively high strengths.

Based on the subsurface conditions encountered at the project site and after site preparation as discussed in this report, the proposed buildings can be supported on conventional shallow spread footing foundation systems bearing on competent natural and/or existing fill Silty Clay or granular engineered fill/lean concrete overlying the competent natural and/or existing fill Silty Clay soils. A maximum allowable bearing pressure of 4,000 psf can be used for the design of the shallow foundations. Competent soils can be identified on the boring logs as Silty Clay exhibiting an unconfined compressive strength value of 1½ tsf or greater or granular soils having a minimum N-value of 8 blows per foot. Following site preparation as recommended, ECS anticipates suitable bearing soils will be available at the anticipated foundation bearing elevations. Because existing fill soils are being recommended for foundation and/or engineered fill support, we recommend that small auger probes and field vane shear or dynamic cone penetrometer tests be performed by ECS at the foundation subgrade during construction to further evaluate the suitability of the bearing soils.

More detailed recommendations with regard to foundations, subgrade preparation and earthwork operations, fill placement, foundation recommendations, slab, loading dock wall and pavement design, underslab drainage, stormwater pond design and construction dewatering are included herein and must be fully reviewed and understood so that the intent of the recommendations are properly utilized during design and construction of the proposed development. We recommend that ECS be retained during construction of the proposed development to monitor all earthwork/subgrade preparation to verify that the exposed subgrade materials and the soil bearing pressures will be suitable for the proposed structure.

Report Prepared By:
Paul J. Giese, P.E.
Principal Engineer

Report Reviewed By:
Brett Gitskin, P.E.
Senior Principal Engineer

PROJECT OVERVIEW

Introduction

This report presents the results of our subsurface exploration and geotechnical engineering analysis performed for the proposed Business Park to be located on the ZPG Nitto Remnant property at the southwest corner of the intersection of 108th Street and Highway 31 in Pleasant Prairie, Kenosha County, Wisconsin. A General Location Diagram, included in the Appendix of this report, shows the approximate location of the project site. Evaluation of environmental considerations for the project was beyond the scope of authorized services for this report.

This study was conducted in general accordance with Proposal No. 42:853-GP (dated August 3, 2016) and authorized by your office via Purchase Order No. CM7010-000057, dated 8/5/2016. In preparing this report, we have utilized information from our current subsurface exploration as well as information from nearby sites.

Site Location

The project site is located at the southwest corner of the intersection of 108th Street and Highway 31, in the Village of Pleasant Prairie, Kenosha County, Wisconsin. The site is bordered by 108th Street to the north, a drive lane for an industrial/office development to the west, a wooded area to the south and Highway 31 to the east.

Existing Site Conditions and Past History

The majority of the site is currently undeveloped and vacant and the southeast portion of the site is used for agricultural purposes (soybeans). Based on review of a *Plat of Survey* (Sheet 1 of 1, dated 05/02/2016 and Revision 1 Soil Borings dated 08/09/2016) prepared by Pinnacle Engineering Group, existing wetlands are located along the western and southwestern portions of the site and extend in an northeast/southwest direction through the southeastern portion of the site (following an existing ditch). In addition, review of a *Concept Site Plan (Version A)* prepared by Pinnacle Engineering Group and dated 06/14/2016 indicates an artificial wetland area (consisting of a former sediment basin) is located in the west-central portion of the site and is partially located within one of the proposed building areas.

Based on information provided by Pinnacle Engineering Group we understand the area of the site to the north and northwest of the wetland/ditch line and to the east of the wetland area along the western property line was previously mass graded and the grading activities were monitored by a geotechnical testing agency. A *Site Grading and Erosion Control Plan* (prepared by STS Consultants, Ltd and dated March 27, 2002) provided by Pinnacle Engineering Group for a portion of the site (east-central along the eastern property line) and which we understand was performed for a previously proposed development, indicates areas where up to about 2½ feet of fill was necessary to achieve final grade. However, no information regarding the limits of fill placement throughout other areas of the site or field reports documenting the placement, compaction and testing of on-site existing fill soils was provided at the time of this report preparation. When and if any information becomes available, we request it be provided to ECS for review and possible revision to the conclusions and recommendations included in this report.

Based on review of topographic information on the *Plat of Survey*, surface elevations across the site are somewhat variable. The existing elevations range between about EL. 702 to EL. 705 (along the western side of the site) to about EL. 728 (at the southeast corner of the site). In general site grades slope from the southeast corner of the site down to the northwest and west. The bottom of the artificial wetland area (former sedimentation basin) is at about EL. 707 and the bottom of the existing ditch through the southeastern portion of the site is at about EL. 709.

Proposed Construction

We understand the proposed construction will consist of two (2) one-story warehouse/manufacturing type buildings. Facility "I" will consist of a 146,000 square foot (sf) (dimensions of 615 feet by 240 feet) structure located in the northeast and central portions of the site. The southwestern end of the building will extend into the area of the artificial wetlands. Facility "II" will consist of a 60,750 sf (dimensions of 385 feet by 160 feet) structure located in the southeast corner (existing soybean field) of the site. We understand the buildings will have 32- to 33-feet tall precast tilt-up walls and will include slabs-on-grade. Facility "I" will have recessed loading docks and truck parking areas along the southeastern side of the building and Facility "II" will have recessed loading docks and truck parking areas along the north side of the building. Automobile parking lots will be located to the north/northwest of Facility "I" and to the west and south of Facility "II". A future employee or semi-trailer parking area may also be located to the southwest of Facility "I".

Structural design information was not available at the time this proposal was written. However, we anticipate the buildings will be supported by a combination of perimeter and interior columns and perimeter wall footings. Based on our conversations with Zilber Property Group, we understand structural loading information was not available at the time of this report; however, based on similar projects, we estimate column and wall loads will be in the range of 100 to 300 kips and 4 to 8 klf, respectively, depending on the proposed structural system.

Based on verbal information provided by Pinnacle Engineering Group, we understand Facility "I" will have a finished first floor elevation (FFE) between EL. 714 and EL. 716. We anticipate the loading docks will be recessed about 4 feet below the FFE. Based on the *Plat of Survey*, existing site grades in the Facility "I" building area range between about EL. 707 (in the former sediment basin) and EL. 719. Therefore, we anticipate up to about 4 to 6 feet of cut and 6 to 8 feet of fill will be required to establish the Facility "I" subgrade elevation, exclusive of site preparation requirements.

Based on verbal information provided by Pinnacle Engineering Group, we understand Facility "II" will have a FFE at EL. 722. We anticipate the loading docks will be recessed about 4 feet below the FFE. Based on the *Plat of Survey*, existing site grades in the Facility "II" building area range between about EL. 726 and EL. 716. Therefore, we anticipate up to about 5 feet of cut and 5 feet of fill will be required to establish the Facility "II" subgrade elevation, exclusive of site preparation requirements.

Four (4) stormwater management ponds are planned for the development. Based on review of the *Concept Site Plan (Version A)* the ponds will be located in the northwest corner, near the southwest corner, along the south-central property line and to the immediate east of Facility "II".

Further information regarding the ponds, including pond bottom elevations, were not known at the time of this report preparation.

If our understanding of the proposed construction is inaccurate, if structural loading information becomes available or if the construction details change, please notify ECS of any changes so that we can determine if the recommendations provided herein are appropriate or if our recommendations need to be modified.

Purpose of Exploration and Scope of Work

The purpose of this exploration was to explore the subsurface conditions at the project site and to develop engineering recommendations to guide the geotechnical design and construction aspects of the project. We accomplished these purposes by performing the following scope of services:

1. Drilling thirteen (13) soil borings within the proposed building areas to a depth of 21 feet below existing site grades using an auger drill rig;

2. Observing and documenting the subsurface conditions within nine (9) backhoe test pits extended to depths ranging from about 4 to 9 feet below existing grades and performed by a backhoe operator (Reesman Excavating) subcontracted by Zilber Property Group on August 4, 2016;

3. Performing laboratory tests on selected representative soil samples from the borings and test pits to evaluate pertinent engineering properties;

4. Analyzing the field and laboratory data from the explorations to develop appropriate engineering recommendations; and,

5. Preparing this geotechnical report of our findings and recommendations.

The conclusions and recommendations contained in this report are based on thirteen (13) soil borings (designated as Borings B-1 through B-13) conducted at the project site under ECS' direction. Borings B-1 through B-7 were drilled within or near the footprint of planned Facility "I" and boring B-8 through B-13 were drilled within or near the footprint of planned Facility "II". Each of the borings was extended to a depth of about 21 feet below the existing surface grade elevations. In addition, the conclusions and recommendations contained in this report are also based on nine (9) backhoe test pits (designated as Test Pits TP-1 through TP-9) extended to depths ranging from about 4 to 9 feet below the existing site grades. Test Pit Nos. TP-1, TP-4, TP-5 and TP-6 were performed within or near the footprint of planned Facility "I" and Test Pit Nos. TP-8 and TP-9 were performed within or near the footprint of planned Facility "II". Test Pit Nos. TP-2, TP-3, TP-7 and TP-9 were performed within or near the planned stormwater pond areas. The test pits were extended to depths ranging from about 4 to 9 feet below the existing site grades.

The soil boring subsurface exploration included split-spoon soil sampling, standard penetration tests (SPT) and groundwater level observations in the boreholes. The backhoe test pit exploration included visual observation of the backhoe excavations as they were being dug, physical measurement of topsoil and encountered soil layers, collection of grab samples of the

various soils encountered and observation and measurement of groundwater conditions. The results of the completed soil borings and backhoe test pits along with a Boring Location Diagram and a Test Pit Location Plan are included in the Appendix of this report.

The borings were located in the field by ECS personnel and the approximate locations are shown on the Boring Location Diagram. The backhoe test pits were performed at random locations throughout the site and the locations of the test pits were determined by Pinnacle Engineering Group after completion of the exploration. Based on our review of the *Plat of Survey* the approximate ground surface elevations at the boring locations ranged between about EL. +707½ and EL. +719 feet in the Facility "I" building area and between about EL. +715 and EL. +725½ feet in the Facility "II" building area. The approximate ground surface elevations at the test pit locations ranged between about EL. +707 and EL. +724 feet. The ground surface elevations are considered to be accurate to within one contour interval (1± foot). The approximate ground surface elevations at the boring locations are shown on the Boring Logs attached in the Appendix of the report. The approximate ground surface elevations at the test pit locations are shown on the Test Pit Summary attached in the Appendix of the report.

EXPLORATION PROCEDURES

Subsurface Exploration Procedures

The soil borings were located in the field by an ECS representative. The ECS subcontracted driller contacted Digger's Hotline, to clear and mark underground utilities in the vicinity of the project site prior to drilling operations. The backhoe test pits were performed at random locations throughout the site as chosen by Zilber Property Group and Reesman Excavating personnel. Reesman Excavating contacted Digger's Hotline, to clear and mark underground utilities in the vicinity of the project site prior to the backhoe test pit operations.

The soil borings were performed with a truck-mounted rotary-type auger drill rig, which utilized continuous flight augers techniques to advance the boreholes. Representative soil samples were typically obtained at 2½-foot intervals in the upper 10 feet, and then 5 foot intervals thereafter to the termination depths of the borings by means of conventional split-barrel sampling procedures. In this procedure, a 2-inch O.D., split-barrel sampler is driven into the soil a distance of 18 inches by a 140-pound hammer falling 30 inches. The number of blows required to drive the sampler through a 12-inch interval, after an initial setting of 6 inches, is termed the Standard Penetration Test (SPT) or N-value and is indicated for each sample on the boring logs. The SPT value can be used as a qualitative indication of the in-place relative density of cohesionless soils. In a less reliable way, it also indicates the consistency of cohesive soils.

The drill rig utilized an automatic trip hammer to drive the sampler. Consideration of the effect of the automatic hammer's efficiency was included in the interpretation of subsurface information for the analyses prepared for this report.

The drill crew maintained a field log of the soils encountered in the borings. After recovery, each geotechnical soil sample was removed from the sampler and visually classified. Representative portions of each soil sample were then sealed in jars and delivered to our laboratory in Brookfield, Wisconsin for further visual classification and laboratory testing. After completion of the drilling operations, the boreholes were backfilled with spoils and patched appropriately at the existing ground surface.

The test pits were backhoe bucket excavated to depths ranging from about 4 to 9 feet below existing site grades. ECS personnel were on-site during excavation of the backhoe test pits to document the subsurface soil and groundwater conditions. Representative grab samples of the various materials encountered were collected, placed in plastic bags and delivered to our laboratory in Brookfield, Wisconsin for further visual classification and laboratory testing. After completion of the backhoe test pit operations, the test pit excavations were backfilled with the excavated soils using nominal compaction effort. Complete removal of the test pit backfill soils and replacement under engineering controlled conditions is recommended in structural areas of the proposed development. If the materials are not removed and backfilled properly, settlement of the test pit backfill soils should be anticipated, along with the resultant distress to structures and pavements.

Laboratory Testing Program

Representative soil samples were selected and tested in our laboratory to check field classifications and to determine pertinent engineering properties. The laboratory testing program included visual classifications, unconfined compressive strength testing of cohesive soil samples utilizing a calibrated pocket penetrometer and moisture content determinations of cohesive soil samples.

Each soil sample was classified on the basis of texture and plasticity in accordance with the Unified Soil Classification System. In addition, the soil samples collected from the backhoe test pits in the vicinity of the planned stormwater ponds were classified in accordance with the USDA Soil Classification System. The group symbols for each soil type are indicated in parentheses following the soil descriptions on the boring logs. A brief explanation of the Unified System is included with this report. The various soil types were grouped into the major zones noted on the boring logs. The stratification lines designating the interfaces between earth materials on the boring logs and profiles are approximate; in-situ, the transitions may be gradual.

Unconfined compressive strength tests were performed on cohesive soil samples from both the test borings and backhoe test pits with the use of a calibrated hand penetrometer. In the hand penetrometer test, the unconfined compressive strength of a soil sample is estimated, to a maximum of 4½ tons per square foot (tsf), by measuring the resistance of a soil sample to penetration of a small, calibrated spring-loaded cylinder.

The soil samples will be retained in our laboratory for a period of 60 days, after which, they will be discarded unless other instructions are received as to their disposal.

EXPLORATION RESULTS

Soil Conditions

The conclusions and recommendations contained in this report are based on twenty (20) soil borings (designated as Borings B-1 through B-20) conducted at the project site under ECS' direction. Borings B-1 through B-20 were drilled within the approximate footprints of the new 3-story apartment structures to depths of about 26 to 36 feet below the existing surface grade elevation. The subsurface conditions encountered at the boring locations performed at the site can be summarized as follows.

Building Borings and Test Pits

The surficial materials encountered at the soil boring and test pit locations consists of topsoil ranging in thickness from about 6 to 36 inches (typically 8 to 16 inches). Below the topsoil, Silty CLAY FILL was encountered at borings B-2 through B-7 and at test pits TP-1, and TP-4 through TP-6 (Facility "I" Building area) to depths ranging from about 2 to 4½ feet below existing grades. No discernable fill soils were encountered at borings B-1 and B-8 through B-13 or at test pits TP-8 and TP-9. At boring B-2, a possible buried topsoil layer (about 6 inches thick) was encountered below the fill at a depth of about 5 feet below existing grade. Native soils underlying the surface materials, fill soils and possible buried topsoil consist of predominantly Silty CLAY and Clayey SILT extending to at least the maximum depth explored, 21 feet. Interbedded SILT, Silty SAND, Sandy SILT or Clayey SAND layers were encountered at borings B-3, B-5 through B-8 and B-11 through B-13. Possible cobbles or boulders were encountered at boring B-8 at about 4 to 5 feet.

The existing FILL soils (Silty CLAY and Clayey SILT) typically exhibited an unconfined compressive strength value (Qp value) greater than 4½ tons per square foot (tsf), indicative of a hard consistency for cohesive materials. The surficial FILL soils were relatively dry to moist at the time of the subsurface explorations with moisture contents in the Silty CLAY and Clayey SILT FILL soils ranging from about 8 to 18 percent (typically 11 to 13 percent). The natural Silty CLAY and Clayey SILT soils exhibited Qp values in the range of 1½ to greater than 4½ tsf, (stiff to hard consistencies) and moisture contents in the range of 11 to 27 percent (typically 12 to 20 percent). The natural SILT, Silty SAND, Sandy SILT or Clayey SAND exhibited SPT N-values ranging from about 10 to 29 blows per foot (bpf) (medium dense).

Detention Pond Test Pits

Test Pit Nos. TP-2, TP-3, TP-7 and TP-9 were performed within or near the planned stormwater detention pond areas. The surface materials at those test pits consisted of about 10 to 36 inches of topsoil (USDA classification of Silty Clay Loam). Existing fill classified as Sandy CLAY (USDA classification of Sandy Clay) was encountered at Test Pit TP-7 to a depth of about 2 ½ feet below the ground surface. Native soils below the fill at Test Pit TP-7 and below the surface topsoil at Test Pit Nos. TP-2, TP-3 and TP-9 consist of predominantly Silty CLAY to Clayey SILT (USDA classification of Silty Clay, Silty Clay Loam and Clay Loam) extending to the maximum depths explored at those locations ranging from about 6 to 8½ feet below existing grades.

General

It should be noted that bid quantity estimation by "averaging" depths and strata changes from boring logs and the test pit summary is not permitted. Too many variations exist for such "averaging" to be valid, particularly in the surficial material thicknesses, soil types and condition, depth and groundwater conditions. A different scope of professional services would be required to obtain subsurface information needed for land purchase considerations. This scope could include additional borings and possibly additional test pits. Even with this additional information, contingencies should always be carried in construction budgets or land purchase agreements to cover variations in subsurface conditions. Soil borings and test pits cannot present the same full-scale view that is obtained during complete site grading, excavation or other aspects of earthwork construction.

Groundwater Observations

Observations for groundwater were made during sampling and upon completion of the drilling operations at the boring locations and during the test pit explorations. Groundwater levels were observed at depths ranging from 7½ to 19½ feet below grade during and upon completion of drilling at test borings B-3, B-5, B-6, B-8, B-9 and B-11 through B-13 and at depths ranging from about 7½ to 9 feet below existing grade at Test Pits TP-1, TP-7 and TP-9 during, upon completion and up to about 2¾ hours after completion of the test pits. No free groundwater was encountered at the remaining boring locations during or upon completion of drilling or the remaining test pit locations.

Glacial till soils in the Midwest frequently oxidize from gray to brown above the level at which the soil remains saturated. The long-term groundwater level is often interpreted to be near this zone of color change. Based on the results of this exploration, the long-term groundwater level may be located at a depth ranging from approximately 7 to 14½ feet below existing site grades which corresponds to approximately between EL. +698½ to EL. +716 feet.

The highest groundwater observations are normally encountered in late winter and early spring and our current groundwater observations are not expected to be at the seasonal maximum water table. It should be noted that the groundwater level can vary based on precipitation, evaporation, surface run-off and other factors not immediately apparent at the time of this exploration. Surface water runoff will be a factor during general construction, and steps should be taken during construction to control surface water runoff and to remove water that may accumulate in the proposed excavations as well as floor slab areas.

ANALYSIS AND RECOMMENDATIONS

Overview

The conclusions and recommendations presented in this report should be incorporated in the design and construction of the project to reduce possible soil and/or foundation related problems.

Based on verbal information provided to us, we understand the existing fill soils encountered across the site were reportedly placed under engineering controlled conditions and tested by a geotechnical engineering company. Based on the previously referenced STS Consultants, Ltd *Site Grading and Erosion Control Plan* we understand the fill soils may have been placed around 2002. The results of the test borings and test pits indicate the encountered fill soils exhibit relatively high strength characteristics which we consider to be representative of soils that may have been placed with compactive effort. The following conclusions and recommendations are provided based on our understanding the existing fill soils consist of engineered fill. When and if any information regarding documentation of the existing fill becomes available, we request it be provided to ECS for review and possible revision to the conclusions and recommendations included in this report.

The following sections present specific recommendations with regard to the geotechnical design and construction aspects of the proposed construction. These include recommendations with regard to subgrade preparation and earthwork, fill placement, building foundations, floor slab design and below-grade walls. Discussion of the factors affecting the building foundations for the proposed construction, as well as additional recommendations regarding geotechnical engineering design and construction aspects at the project site, are included below. We recommend that ECS review the final design and specifications to check that the earthwork and foundation recommendations presented in this report have been properly interpreted and implemented in the design and specifications.

Subgrade Preparation and Earthwork Operations

Initial Subgrade Preparation

Initial subgrade preparation for the new building pads and pavement areas should consist of complete removal of existing topsoil, trees, brush, root bulbs and other deleterious organic or refuse material. Based on the conditions at the test boring and test pit locations, topsoil depths ranged from about 6 to 36 inches (typically 8 to 16 inches). However, considering the presence of the existing wetlands, the existing ditch area and the artificial wetland area (former sediment basin) areas of deeper removal may be necessary in these and other unexplored areas of the site. The topsoil can also be stockpiled at the subject property for potential reuse in landscaping and other non-structural applications.

Once the surficial materials have been removed, the limits of the structure footprints and proposed pavement areas should be excavated to the design subgrade elevation and/or the planned new fill areas be properly prepared. ECS does not recommend the slab/pavement subgrades remain exposed to the elements or construction traffic for a prolonged period of time as the predominant Silty CLAY and Clayey SILT subgrade may be disturbed and/or softened. If

the slabs/pavements are not planned to be constructed or engineered fill is not planned to be placed within a few days after exposing the final design or structural fill area subgrades, consideration should be given to leaving the subgrade approximately 1 foot above the final design subgrade to help prevent softening of the design subgrade soils (if feasible).

Based on verbal information provided by Pinnacle Engineering Group, we understand Facility “I” will have a FFE between EL. +714 and EL. +716 and Facility “II” will have a FFE at EL. 722. Therefore, based on the existing site grades, we anticipate up to about 4 to 6 feet of cut and 6 to 8 feet of fill will be required to establish the Facility “I” subgrade elevation and about 5 feet of cut and 5 feet of fill will be required to establish the Facility “II” subgrade elevation, exclusive of site preparation requirements.

Based on the conditions encountered at the Facility II boring and test pit locations, we expected the subgrades will consist of predominantly native Silty CLAY. Based on the conditions encountered at the Facility “I” boring locations, following the above recommended initial site preparation ECS expects the exposed subgrades will typically consist of a combination of existing Silty CLAY FILL and natural Silty CLAY. Based on the results of the subsurface exploration and the verbal information that existing fill soils on the site were reportedly placed under engineering controlled conditions, we do not recommend complete removal of the existing fill soils. However, prior to any new fill placement or building or pavement construction we do recommend complete evaluation of the existing fill soils as described in the following paragraphs.

Once the subgrade soils have been exposed, and prior to any new fill placement, the subgrades should be proofrolled using a loaded dump truck having an axle weight of at least 10 tons. The intent of the proofroll is to aid in identifying localized soft or unsuitable material which may be required to be removed. If soft or yielding soils are observed during the proofroll of the subgrade, the soft soils should be undercut up to a maximum of 1 foot (in the pavement areas) or 2 feet (in the building slab footprint) and replaced with compacted engineered fill to the design subgrade. The final depth of undercutting, if any, should be determined by ECS based upon the results of the construction phase evaluations. In the area of Facility “I” proofrolling will only identify near-surface soils that are unsuitable for pavement and slab support. Any deeper pockets of unsuitable FILL, or other deleterious soils may not be fully identified, which could lead to premature deterioration/cracking of the building slab and pavements. Therefore, leaving the existing FILL in-place and filling above existing FILL must include the acceptance of a higher risk of premature deterioration/cracking of the pavements and slab when compared to complete removal/replacement of existing FILL.

To help limit the volume of soil removed as a result of the proofrolling and/or testing observations, we recommend soft or yielding soils be evaluated in approximately 6 to 12-inch intervals. That is to say, if soft or yielding soils are identified, the contractor should remove 6 to 12 inches of material in the subject area and then proofroll/evaluate the undercut subgrade. This could potentially limit the need to remove 2 feet of soil at once at all locations where soft or yielding soils are identified. A DCP (dynamic cone penetrometer) or Army Corps of Engineers penetrometer can also be used in conjunction with proofrolling to establish appropriate depths for remedial action.

At the time of the subsurface explorations, the surficial Silty CLAY existing FILL and native soils were considered to be in a relatively dry condition. Therefore, prior to any new fill placement, we

recommend the exposed soils be scarified and moisture conditioned to ±2 percentage points of the optimum moisture content. The exposed soils should then be compacted to a minimum of 95 percent of the maximum dry density obtained in accordance with ASTM Specification D-1557, Modified Proctor Method.

General

When feasible, ECS recommends that the subgrade recommendations are employed 10 lateral feet beyond the limits of the building footprints and 5 lateral feet beyond the limits of new pavement areas. Steps should be taken by the contractor to control surface water runoff and to remove water from precipitation that may accumulate in the subgrade areas, especially during the wet season. When wet and subjected to construction traffic, softening and disturbance of the exposed subgrade may occur. Construction traffic should be limited when the subgrade is wet. During final preparation of the subgrade, a smooth drum roller is often used to provide a flat surface and provide for better drainage to reduce the negative impact of rain events. We also recommend crowning or sloping the subgrade to provide positive drainage of the subgrades. Care should be exercised during subgrade preparation (i.e. compaction operations) to reduce pumping of the subgrade soils and observations should be made during earthwork operations to determine whether the weight of the construction equipment is negatively impacting the on-site soils. Minimizing disturbance to the exposed subgrade is important for proper construction of the slabs-on-grade.

Exposure to the environment may weaken the subgrade soils if the excavations remain open for too long a period. If the subgrade soils are softened by surface water intrusion or exposure, the softened soils must be removed from the subgrade excavation bottom immediately prior to placement of concrete and/or engineered fill. We recommend the foundation excavations be filled with concrete the day they are excavated, or if not feasible, the subgrade protected with a crushed stone or lean concrete working mat, if the excavations will be left open overnight.

Excavations should comply with the requirements of OSHA 29CFR, Part 1926, Subpart P, "Excavations" and its appendices, as well as other applicable codes. This document states that the contractor is solely responsible for the design and construction of stable, temporary excavations. The excavations should not only be in accordance with current OSHA excavation and trench safety standards but also with applicable local, state, and federal regulations. The contractor should shore, slope or bench the excavation sides when appropriate. Due to the depths of excavations, especially along the existing roadways, there is the potential to undermine adjacent roadways and utilities. The contractor should use care as to not undermine the existing adjacent roadways and utilities and incorporate underpinning or other support methods as appropriate.

If problems are encountered during the earthwork operations, or if site conditions deviate from those encountered during our subsurface exploration, ECS should be notified immediately. We recommend that the project geotechnical engineer or his representative should be on site to monitor stripping and site preparation operations and observe that unsuitable soils have been satisfactorily removed and observe the proofrolling of the subgrades.

Fill Placement

All fills should consist of an approved material, free of organic matter and debris, particles greater than 3-inches and have a Liquid Limit and Plasticity Index less than 40 and 15, respectively. Unacceptable fill materials include topsoil and organic materials (OH, OL), high plasticity silts and clays (CH, MH), and low-plasticity silts (ML). Under no circumstances should high plasticity soils be used as fill material in proposed structural areas or close to site slopes. The soils encountered across the site could be reutilized as engineered fill after removal of deleterious materials (such as buried topsoil) or debris greater than 3 inches in diameter. We do not recommend utilizing pea gravel as engineered fill as these materials result in large voids between particles and the have the potential for long-term settlement.

The on-site soils (primarily the near surface natural and existing fill Silty CLAY) will likely require moisture content adjustments in addition to screening, such as the application of discing or other drying techniques or spraying of water to the soils prior to their use as compacted fill (termed manipulation). For onsite soils found suitable (after ECS' evaluation during construction) for use as engineered fill, the contractor should be prepared to implement discing or other drying techniques (termed manipulation) and recognize and account for increased costs associated with manipulation of the suitable onsite soils. **In addition, if construction occurs in late fall to early spring, drying the onsite cohesive soils may not be possible and the contractor should budget for offsite fill, if required.**

Fill materials should be placed in lifts not exceeding 8-inches in loose thickness and moisture conditioned to within ±2 percentage points of the optimum moisture content. Soil bridging lifts should not be used, since excessive settlement of overlying structures will likely occur. Controlled fill soils should be compacted to a minimum of 95 percent of the maximum dry density obtained in accordance with ASTM Specification D-1557, Modified Proctor Method.

The expanded footprint of the proposed building and fill areas should be well defined, including the limits of the fill zones at the time of fill placement. Grade control should be maintained throughout the fill placement operations. All fill operations should be observed on a full-time basis by a qualified soil technician to determine that the specified compaction requirements are being met. A minimum of one compaction test per 2,500 square foot area or 50 linear feet of trench should be tested in each lift placed, or minimum of 3 tests per lift. The elevation and location of the tests should be clearly identified at the time of fill placement.

Compaction equipment suitable to the soil type used as fill should be used to compact the fill material. Theoretically, any equipment type can be used as long as the required density is achieved; however, the standard of practice typically dictates that a vibratory roller is utilized for compaction of granular soils and a sheepsfoot roller be utilized for compaction of cohesive soils. In addition, a steel drum roller is typically most efficient for compacting and sealing the surface soils. All areas receiving fill should be graded to facilitate positive drainage from building pad areas of free water associated with precipitation and surface runoff.

It should be noted that prior to the commencement of fill operations and/or utilization of off-site borrow materials, the Geotechnical Engineer of Record should be provided with representative samples to determine the material's suitability for use in a controlled compacted fill and to develop moisture-density relationships. To expedite the earthwork operations, if off-site borrow materials are required, it is recommended they consist of suitable fill materials in accordance

with the recommendations previously outlined in this section. If frost susceptible soils are imported to the project site, the frost susceptible soils should not be placed within 3½ of final site grades.

Fill materials should not be placed on frozen soils or frost-heaved soils and/or soils that have been recently subjected to precipitation. All frozen soils should be removed prior to continuation of fill operations. Borrow fill materials, if required, should not contain frozen materials at the time of placement. All frost-heaved soils should be removed prior to placement of controlled, compacted fill, granular subbase materials, and foundation or slab concrete. Pea gravel should not be used as engineered fill.

Foundation Recommendations

The proposed structures can be supported on shallow foundation systems (i.e., wall and spread footings) bearing in a combination of competent natural soils and new engineered fill overlying competent existing (natural and fill) soils. Shallow foundation systems bearing in the competent natural soils, existing fill or new engineered fill/lean concrete overlying competent existing (natural and fill) soils can be designed for a maximum allowable soil bearing pressure of 4,000 psf. Competent soils can be identified on the boring logs as Silty Clay exhibiting an unconfined compressive strength value of 1½ tsf or greater or granular soils having a minimum N-value of 8 blows per foot. After site preparation as recommended in this report, we anticipate suitable bearing soils will be available at the planned foundation bearing grades. However, considering the past filling operations that have occurred and the potential for the presence of buried topsoil (as encountered at boring B-2) some areas of undercutting may be necessary and should be anticipated.

If excavations are performed next to existing roadways or utilities, the contractor should be cautious as not to damage the existing roadways or utilities, or subgrade soils. The contractor should avoid disturbing soils located at a 1:1 slope (H:V) from the bottom/outward edge of existing structures. Engineered supports (i.e., underpinning and/or earth retention systems) are required if the excavations could potentially undermine the existing footings roadways or utilities.

General - Shallow Foundations

The engineered fill should be compacted to a minimum of 95 percent of the maximum dry density in accordance with Modified Proctor Method, ASTM Specification D 1557. The zone of the engineered fill placed below the foundations should extend 1 foot beyond the outside edges of the footings and from that point, outward laterally 1 foot for every 2 feet of fill thickness below the footing. If lean concrete is used to raise the footing subgrades, lateral over-excavation is typically not required but the lean concrete should be 1 foot wider than the footing dimensions (i.e., 6 inches beyond the outer edges of the footings).

We recommend that the excavation/backfill of foundations be monitored full-time by an ECS Geotechnical Engineer or his representative to verify that the soil bearing pressure is consistent with the boring log information obtained during the geotechnical exploration. **Due to the presence of existing fill being present below the bearing depth, we recommend that small auger probes and field vane shear or dynamic cone penetrometer tests be**

performed to at least half the width of the footing, or a minimum of 3 feet below each isolated column footing and to at least 2 feet below continuous footings. At least one auger probe should be performed at each column footing and probes should be performed at 25 foot intervals along continuous footings to verify the suitability of the soils to support the recommended maximum net allowable bearing pressure or unsuitable soils is present below the foundations.

To help reduce the potential for foundation bearing failure and excessive settlement due to local shear or "punching" action, we recommend that continuous footings have a minimum width of 18 inches and that isolated column footings have a minimum lateral dimension of 30 inches. In addition, footings should be placed at a depth to provide adequate frost cover protection. For this region, we recommend the exterior footings and footings beneath unheated areas be placed at a minimum depth of 4 feet below finished grade. Interior footings can be placed at a minimum depth of 2 feet below grade provided that suitable soils are encountered and that the foundations will not be subjected to freezing weather either during or after construction.

Settlement of individual footings, designed in accordance with our recommendations presented in this report, is expected to be small and within tolerable limits for the proposed building. For footings placed on suitable natural soils or properly compacted engineered fill, maximum total settlement is expected to be in the range of 1 inch or less. Maximum differential settlement between adjacent columns is expected to be in the range of ½ inch. These settlement values are based on our engineering experience with the soil and the anticipated structural loading, and are to guide the structural engineer with his/her design.

Floor Slab Design

For the design and construction of the slabs-on-grade for the buildings, the recommendations provided in the section entitled **Subgrade Preparation and Earthwork Operations** should be followed. The building floor slab thicknesses can be determined utilizing an assumed modulus of subgrade reaction of 150 pounds per cubic inch (pci). We recommend the slabs be designed with a minimum thickness of 4 inches in office areas and a minimum of 6 inches in warehouse/loading dock areas.

We recommend the slabs-on-grade be underlain by a minimum of 6 inches of granular material having a maximum aggregate size of 1½ inches and no more than 5 percent of fines (i.e., WisDOT Section 310 Base Course or similar). This granular layer will facilitate the fine grading of the subgrade and help prevent the rise of water through the floor slabs. Prior to placing the granular material, the floor subgrades should be free of standing water, mud, and frozen soil. Before the placement of concrete, a vapor barrier may be placed on top of the granular material to provide additional moisture protection. Welded-wire mesh reinforcement should be placed in the upper half of the floor slab and attention should be given to the surface curing of the slabs to minimize uneven drying of the slabs and associated cracking and/or slab curling. The use of a blotter or cushion layer above the vapor retarder can also be considered for project specific reasons. Please refer to ACI 302.1R04 *Guide for Concrete Floor and Slab Construction* and ASTM E 1643 *Standard Practice for Installation of Water Vapor Retarders Used in Contact with Earth or Granular Fill Under Concrete Slabs* for additional guidance on this issue.

We recommend that the floor slabs be isolated from the foundation footings so differential settlement of the structure will not induce shear stresses on the floor slab. For maximum effectiveness, temperature and shrinkage reinforcements in slabs on ground should be positioned in the upper third of the slab thickness. The Wire Reinforcement Institute recommends the mesh reinforcement be placed 2 inches below the slab surface or upper one-third of slab thickness, whichever is closer to the surface. Adequate construction joints, contraction joints and isolation joints should also be provided in the slab to reduce the impacts of cracking and shrinkage. Please refer to ACI 302.1R04 Guide for Concrete Floor and Slab Construction for additional information regarding concrete slab joint design.

Underslab Drainage Considerations

Based on the results of this exploration, the long-term groundwater level may be located at a depth ranging from approximately 7 to 9½ feet below existing site grades in the area of Facility "I" which corresponds to approximately between EL. +699 to EL. +710 feet and at a depth of about 7 to 8½ feet below existing site grades in the area of Facility "II" which corresponds to approximately between EL. +703½ to EL. +716 feet. The FFE at Facility "I" is planned to be at EL. +714 to EL. +716, with a recessed dock anticipated to be about 4 feet lower or between EL. +710 to EL. +712. The FFE at Facility "II" is planned to be at EL. +722, with a recessed dock anticipated to be about 4 feet lower or at about EL. +718.

We do not anticipate a significant volume of water will persist at floor slab-on-grade or loading dock slab elevations based on the conditions encountered. It should be noted however that surface runoff and limited groundwater seepage may accumulate at the slab subgrades such as a perched water condition or due to seasonal water level variations. The contractor should be prepared to deal with water in case it becomes an issue. As such, we recommend that positive drainage be implemented around the perimeter of the proposed structures (e.g., the surface grades should slope down from the building wall) to reduce the potential for water accumulation under the floor slab and foundation elements, which could potentially weaken the bearing soils. We recommend that floor slabs be underlain by a minimum of 6 inches of granular layer to help prevent the rise of water through the floor slabs as recommended in **Floor Slab Design** section. Building downspout should be installed such that flow water is directed away from the building walls.

Below-Grade Loading Dock Walls

Lateral Earth Pressures

Permanent below-grade walls, such as the loading dock retaining walls, should be designed to withstand lateral earth pressures and surcharge loads. The lateral earth pressures exerted on the walls will be a function of the stiffness and the rotation of the walls. The rotation of the wall controls the degree to which the internal strength of the soil is mobilized. If rotation or deflection of the walls will be less than that required to mobilize the active earth pressure condition due to stiffness, bracing or other mechanism (as is typical with basement walls), the "at-rest" earth pressure condition should be evaluated. For the at-rest earth pressure condition, below grade walls can be designed for a linearly increasing lateral earth pressure of 65 psf per vertical foot of wall. The at-rest earth pressure of 65 psf per vertical foot of wall assumes that the below-

grade walls will be in a drained condition (i.e., no hydrostatic forces on the back of the wall). In the event that the walls remain undrained, a linearly increasing lateral earth pressure of 65 psf per vertical foot of wall should be utilized above the long-term groundwater level and 100 psf per vertical foot of wall should be utilized below the long-term groundwater level. A long-term groundwater level of approximately EL. +710 in the Facility "I" building area and a long-term groundwater level of approximately EL. +716 in the Facility "II" building area is recommended for design. The wall design should also account for surcharge loads within a 45 degree slope from the base of the wall, including floor loads near the loading docks. For the at-rest earth pressure condition, a lateral earth pressure coefficient of 0.45 should be applied to surcharge loads.

The "active" earth pressure condition, such as for retaining walls not associated with structures, which results in the minimum applied earth pressure, results when the rotation of the wall about its base and away from the retained soil is approximately 0.001 times the height of the wall or greater. This is typically the case with cantilever-type walls. If the active earth pressure condition develops, we recommend below-grade walls be designed for a linearly increasing lateral earth pressure of 40 psf per vertical foot of wall above the long-term groundwater level and 85 psf/ft below the long-term groundwater level. These active lateral earth pressures assume that granular materials are used for wall backfill. The wall design should also account for surcharge loads within a 45 degree slope from the base of the wall. For the active earth pressure condition, a lateral earth pressure coefficient of 0.33 should be applied to surcharge loads.

To develop "passive earth" pressures, the footings should be located at depths greater than the depth of frost penetration (4 feet below lowest adjacent finished grade). Passive resistance within the zone of seasonal volume change (i.e., approximately 4 feet from the final ground surface) should be neglected. For footings and walls located below the frost penetration of 4 feet (if any) and if the passive condition develops, a passive lateral earth pressure of 360 psf per vertical foot of wall can be used to design below grade walls for granular backfill. As stated, the upper 4 feet of soil should be neglected when determining the passive resistance of the soil.

Below-Grade Wall Backfill

The space between the outside of the walls and the excavation should be backfilled with a granular fill extending to a level of approximately 2 feet below the final outside grade. The remaining 2 feet should consist of a clayey material to minimize the amount of surface water infiltration into the granular material, and thus, reduce the excess water to be handled by the drainage system. Asphaltic concrete or Portland cement concrete can also be used to cover the ground surface and minimize the surface water infiltration. The ground surface adjacent to the below-grade walls should be kept properly graded to prevent ponding of water adjacent to the below-grade walls.

Special attention should be employed during placement of new fill against below-grade walls. Based on our experience, fill soils placed against below-grade walls and immediate areas along Below-grade walls are often not compacted/densified adequately due to space constraints and lack of compaction effort. Inadequate placement and compaction of new fill will result on at-grade slab/pavement subgrade settlement and distress. A jumping jack, walk-behind vibratory plate or similar equipment should be used in manhole areas or other limited areas that are not feasible with a heavy-duty pneumatic tire or smooth drum roller. To achieve a desirable balance

between minimizing excessive pressures against the below grade walls and reducing the settlement of the wall backfill, we recommend that the wall granular backfill be compacted to at least 90 percent of the maximum dry density obtained in accordance with ASTM Specification D 1557, Modified Proctor Method. Thinner lifts of 4 to 6 inches can be effective in improving compaction performance. Where the fill materials will be supporting sidewalks or pavements, the upper 2 feet should be compacted to 95% of the maximum dry density referenced above.

Backfill materials should consist of inorganic materials, free of debris, be free draining, and containing no frost susceptible soil. The fill placed adjacent to the below grade walls should not be overcompacted. Heavy earthwork equipment should maintain a minimum horizontal distance away from the below-grade walls of 1 foot per foot of vertical wall height. Lighter compaction equipment should be used close to the below grade walls. Where light (e.g., hand) compaction equipment is employed, the maximum lift thickness should be reduced to 6 inches.

Suitable man-made drainage materials may be used in lieu of the granular backfill, adjacent to the below-grade walls. Examples of suitable materials include Enka Mat, Mira Drain, or geotec Drains. If the excavation support system is used as a back form for wall construction, the geosynthetic drainage media can be placed directly against the support system prior to placing reinforcing steel. These materials should be covered with a filter fabric having an apparent opening size (AOS) consistent with the size of the soil to be retained (or with cast-in-place concrete). The material should be placed in accordance with the manufacturer's recommendations and connected to a perimeter drainage system, which in turn should be properly drained.

Below-Grade Wall Perimeter Drainage

We recommend loading dock below-grade walls be provided with a perimeter drainage system to reduce the potential for excess hydrostatic pressures to be exerted on the walls. This system may consist of perforated or porous wall, closed joint drain tiles located around the perimeter of the walls, at the footing level. These drain lines should be surrounded by a minimum of 8 inches of free draining granular filter material having a gradation compatible with the size of the openings utilized in the drain lines and surrounding soils to be retained. The drainage tiles should be connected to a permanent sump with automatic and stand-by pump arrangement, or to a storm sewer, to remove water which may accumulate. Occupied below-grade space should include dampproofing, waterproofing and waterstops, as required by local building code requirements.

Pavement Design Recommendations

We recommend that the pavement subgrade be prepared in accordance with the **Subgrade Preparation and Earthwork Operations** section of this report. Once the subgrade has been properly prepared, we recommend the minimum pavement sections detailed in Table 1 for the proposed development. The recommended pavement sections were developed based on the anticipated traffic loads, assuming a California Bearing Ratio (CBR) of 3 for the predominant Silty CLAY FILL and native subgrade soils which are considered to provide the least pavement support and our local experience. All pavement materials and construction should be in accordance with the Guidelines for AASHTO Pavement Design and Wisconsin Department of Transportation (WisDOT) Standards for Road and Bridge Construction.

Table 1: Pavement Section Recommendations

Pavement Material	Compacted Material Thicknesses (Inches)			
	Flexible Pavement (Light Duty)	Flexible Pavement (Heavy Duty)	Rigid Pavement (Light Duty)	Rigid Pavement (Heavy Duty)
Portland Cement Concrete	--	--	5	8
Bituminous Surface Course	1½	2	--	--
Bituminous Base Course	2	3	--	--
Crushed Granular Subbase	8	12	6	6
Total Pavement Section Thickness	11½	17	11	14

The pavement sections specified in the table above are general pavement recommendations based on the anticipated usage at the project site and were not developed based on specific traffic patterns/loading and resiliency factors, as those parameters were not provided by the design team. We recommend the project team provide ECS with actual design traffic loads, if greater than the assumed traffic loads discussed below, so that we can verify the recommendations detailed herein are appropriate for the anticipated traffic loads. The table above provides "Light Duty" and "Heavy Duty" flexible and rigid pavement recommendations. The light-duty pavement section assumes that typical traffic loading will be limited to standard automobiles and does not account for more heavily loaded vehicles (i.e., multiple axle trucks) and should be used for parking lanes. The "Heavy-Duty" pavement section is recommended for pavements to be subjected with frequent traffic such as drive lanes, the loading dock aprons and points of ingress/egress and a maximum traffic load of 50 semi-tractor trailers per day. If the anticipated traffic will exceed 50 semi-tractor trailers per day, ECS should be contacted for a revised heavy duty pavement design, otherwise, a shortened pavement life should be expected. The rigid pavement section could be considered for loading areas, truck docks, trailer staging areas (at least for the trailer pads) and dumpster enclosure areas.

It should also be noted that the pavement sections specified in the table above were developed for the anticipated in-service traffic conditions only and do not provide an allowance for construction traffic conditions or traffic conditions in excess of typical residential/collector street traffic. Therefore, if pavements will be constructed early during site development to accommodate construction traffic, consideration should be given to the construction of designated haul roads, where thickened pavement sections can be provided to accommodate the construction traffic, as well as the future in-service traffic. ECS can provide additional design assistance with pavement sections for haul roads upon request.

We recommend the crushed granular base course should be compacted to at least 95 percent of the maximum dry density obtained in accordance with ASTM D1557, Modified Proctor method. During asphalt pavement construction, the wearing and leveling course should be compacted to a minimum of 93 percent of the theoretical density value. Prior to placing the

granular material, the pavement subgrade soil should be properly compacted, observed to be stable during a final proofroll and free of standing water, mud, and frozen soil.

Adequate construction joints, contraction joints and isolation joints should be provided in the areas of rigid pavement to reduce the impacts of cracking and shrinkage. Please refer to ACI 330R-92 *Guide for Design of Concrete Parking Lots*. The Guide recommends an appropriate spacing strategy for the anticipated loads and pavement thickness. It has been our experience that joint spacing closer to the minimum values results in a pavement with less cracking and better long term performance.

The pavements should be designed and constructed with adequate surface and subsurface drainage. Where standing water develops, either on the pavement surface or within the base course layer, softening of the subgrade and other problems related to the premature deterioration of the pavement can be expected. Furthermore, good drainage should minimize the possibility of the subgrade materials beneath the pavement becoming saturated over a long period of time. Infiltration and subterranean water are the two sources of water that should be considered in the pavement design for the project. Infiltration is surface water that enters the pavement through the joints, pores, cracks in the pavement and through shoulders and adjacent areas pavements as a result of precipitation. Subterranean water is a source of water from a high water table on the site. The long term groundwater level on the site is estimated to be located at depths ranging from 7 to 14½ feet below existing site grades. Therefore, infiltration is the most important source of water to be considered for this project.

Pavement Maintenance

Because existing FILL soils are being considered to remain in place below the planned pavements, an increased risk of pavement distress must be accepted and we strongly recommend regular maintenance of the pavement structure be performed. Regular maintenance and occasional repairs should be implemented to keep pavements in a serviceable condition. In addition, to help minimize water infiltration to the pavement section and within the base course layer resulting in softening of the subgrade and deterioration of the pavement, we recommend the timely sealing of joints and cracks using elastomeric caulk in existing pavement. We recommend exterior pavements should be reviewed for distress/cracks twice a year, once in the spring and once in the fall.

Sound maintenance programs should help maintain and enhance the performance of pavements and attain the design service life. A preventative maintenance program should be implemented early in the pavement life to be effective. The “standard in the industry” supported by research indicates that preventative maintenance should begin within 2 to 5 years of the placement of pavement. Failure to perform preventative maintenance will reduce the service life of the pavement and increase the costs for both corrective maintenance and full pavement rehabilitation.

Detention Pond Considerations

The soils encountered at the test pits within and near the planned detention ponds consist of Silty Clay Loam and Silty Clay materials. These soils have low infiltration rates estimated at

0.07 inches per hour and 0.04 inches per hour, based on the rates included in Table 2 of the Wisconsin Department of Natural Resources (WDNR) *Site Evaluation for Stormwater Infiltration* (1002) document. Based on the encountered soils, the site is considered to be exempt from infiltration requirements in accordance with NR 151.12(5)(c)6.a of the Wisconsin Administrative Code. A copy of the WDNR *Soil Evaluation – Storm* form for the site is included in the Appendix of this report.

We recommend ECS and the project team evaluate the soils in the vicinity of the final detention pond locations at the time of construction if soils other than Silty Clay are encountered. A low permeability clay liner will most likely need to be constructed on the slopes and bottom of the pond. The liner should be constructed using on site lean clay soils or clean imported clay soils. The fill should be placed in lifts not exceeding 8 inches in loose thickness, moisture conditioned to within 2 percent of the optimum moisture content and compacted to at least 90 percent of the maximum dry density obtained in accordance with ASTM D 1557. The encountered clayey soils encountered at the site generally appear suitable for liner material provided they include relatively small amounts of sand and silt. We recommend the suitability of onsite and offsite clay soils for liners should be evaluated and tested prior to their use. Some sorting of materials that are unsuitable for use as clay liner may be necessary considering the presence of fill soils. The project civil engineer must provide specific design recommendations regarding the detention pond.

For construction of the clay liners, it may be necessary to bench the side walls of the pond horizontally, with 1 to 3-foot vertical steps. This would allow horizontal placement and compaction of the liner section. However, adequate compaction for the purpose of detention is probably possible for cohesive fill placed in lifts parallel to the cut slope. The exposed cut surfaces at the pond excavation sides and bottom may tend to become disturbed during excavation process. We recommend a minimum 12 inches below the finish grade be scarified and recompacted to a minimum of 90 percent of Modified Proctor maximum dry density value.

To minimize erosion of the sides of the basin excavation and embankments, erosion protection should be provided. The placement of a granular rip-rap and/or establishment of vegetation can be considered for erosion control within the pond and surrounding areas. A slope of 3:1 (Horizontal:Vertical) or flatter should be used to reduce the potential for slope stability related problems within side slopes of the pond.

PROJECT CONSTRUCTION RECOMMENDATIONS

General Construction Considerations

We recommend that the subgrade preparation, installation of the foundations and construction of slabs be monitored by an ECS geotechnical engineer or his representative. Methods of verification and identification such as proofrolling, DCP testing, vane shear tests, and hand auger probe holes will be necessary to further evaluate the subgrade soils and identify unsuitable soils. The contractor should be prepared to over-excavate footing and slab-on-grade excavations at isolated locations (as necessary). We recommend that excavations of new foundations be monitored on a full-time basis by an ECS geotechnical engineer or his representative to verify that the soil bearing pressure and the exposed subgrade materials will be suitable for the proposed structures and are consistent with the boring log information obtained during this geotechnical exploration. We would be pleased to provide these services.

Due to the presence of existing fill soils and the possibility of unsuitable soils (i.e. buried topsoil) being present below the foundation bearing depths, we recommend that small auger probes and field vane shear or dynamic cone penetrometer tests be performed to at least half the width of the footing, or a minimum of 3 feet below each isolated column footing and to at least 2 feet below continuous footings. At least one auger probe should be performed at each column footing and probes should be performed at 25 foot intervals along continuous footings to verify the suitability of the soils to support the recommended maximum net allowable bearing pressure or unsuitable soils is present below the foundations.

Construction Dewatering

Based on the groundwater conditions encountered at the project site, we do not anticipate that significant dewatering efforts will be necessary during site preparation and construction. Minor groundwater seepage and/or surface runoff may introduce water into the project site and the general contractor should be prepared to remove accumulated water prior to the placement of fill and concrete. We anticipate that the removal of accumulated water can be achieved utilizing drainage trenches and a sump and pump system for shallow areas, however, specialized dewatering techniques may be required where water bearing Sand zones are encountered.

Exposure to the environment may weaken the soils at the footing bearing levels if the foundation excavations remain open for too long a period. If the bearing soils are softened by surface water intrusion or exposure, the softened soils must be removed from the foundation excavation bottom immediately prior to placement of concrete.

Closing

This report has been prepared in order to aid in the evaluation of this property and to assist the architect and/or engineer in the design of this project. The scope is limited to the specific project and locations described herein and our description of the project represents our understanding of the significant aspects relative to soil and foundation characteristics. In the event that any change in the nature or location of the proposed construction outlined in this report are planned, we should be informed so that the changes can be reviewed and the

conclusions of this report modified or approved in writing by the geotechnical engineer. It is recommended that all construction operations dealing with earthwork and foundations be reviewed by an experienced geotechnical engineer to provide information on which to base a decision as to whether the design requirements are fulfilled in the actual construction. If you wish, we would welcome the opportunity to provide field construction services for you during construction.

The analysis and recommendations submitted in this report are based upon the data obtained from the soil borings and tests performed at the locations as indicated on the Boring Location Diagram and other information referenced in this report. This report does not reflect any variations, which may occur between the borings. In the performance of the subsurface exploration, specific information is obtained at specific locations at specific times. However, it is a well-known fact that variations in soil conditions exist on most sites between boring locations and also such situations as groundwater levels vary from time to time. The nature and extent of variations may not become evident until the course of construction. If variations then appear evident, after performing on-site observations during the construction period and noting characteristics and variations, a reevaluation of the recommendations for this report will be necessary.

In addition to geotechnical engineering services, ECS Midwest, LLC has the in-house capability to perform multiple additional services as this project moves forward. These services include the following:

- Seismic Site Classification Testing
- Environmental Consulting;
- Project Drawing and Specification Review; and,
- Construction Material Testing / Special Inspections

We would be pleased to provide these services for you. If you have questions with regard to this information or need further assistance during the design and construction of the project please feel free to contact us.

APPENDIX

Site Location Diagram

Boring Location Diagram

Boring Logs

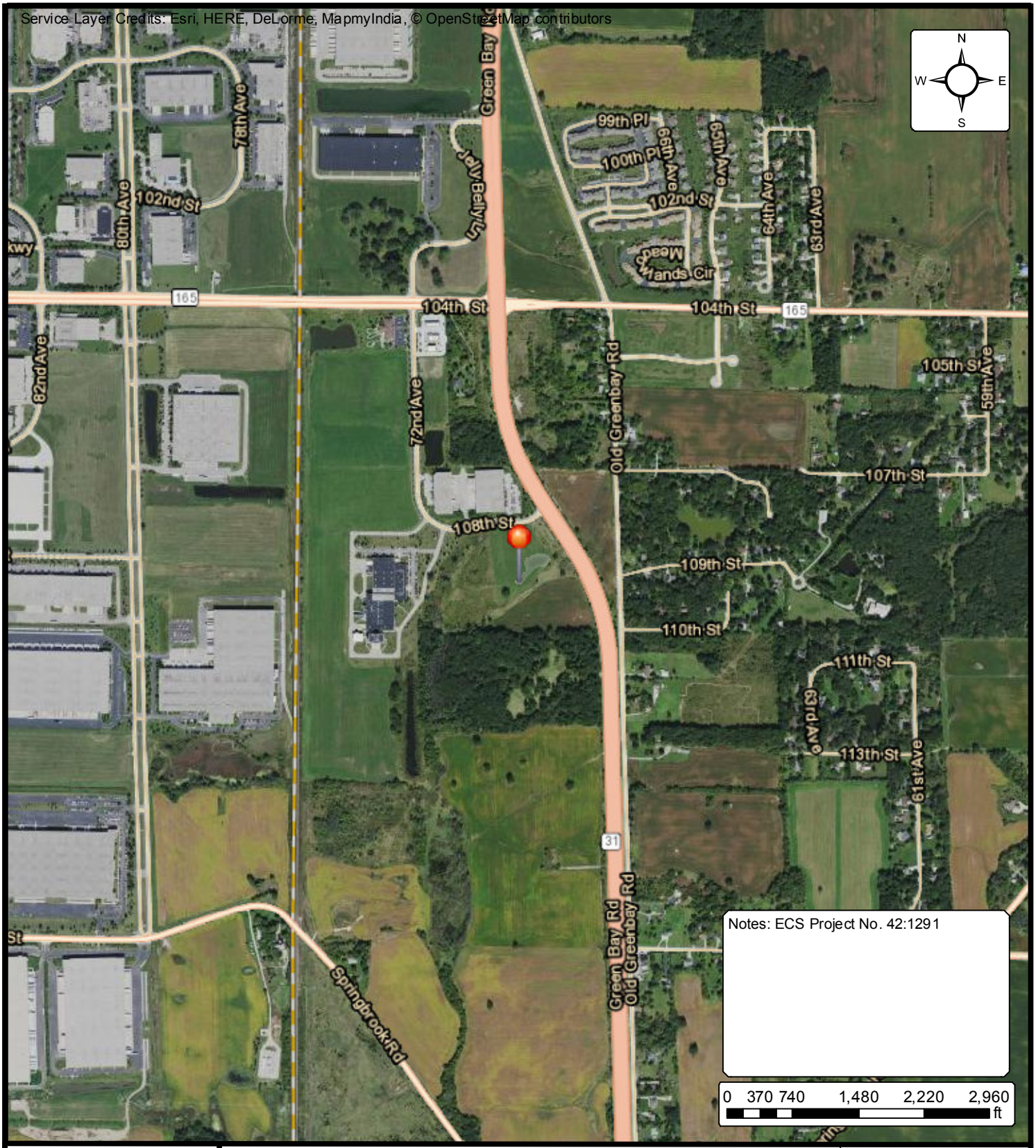
Test Pit Location Plan

Test Pit Summary

WDNR Soil Evaluation Storm Form

Unified Soil Classification System

Reference Notes for Boring Logs



Site Location Diagram

ZPG NITTO DENKO REMNANT PROPERTY

ZILBER PROPERTY GROUP
PLEASANT PRAIRIE WI

ENGINEER	PJG
SCALE	1 " = 1,500 '
PROJECT NO.	
SHEET	1 OF 1
DATE	8/25/2016



Boring Location Diagram

ZILBER PROPERTY GROUP

ZPG NITTO DENKO REMNANT PROPERTY

PLEASANT PRAIRIE WI

ENGINEER	PJG
SCALE	1 " = 150 '
PROJECT NO.	
SHEET	1 OF 1
DATE	8/22/2016

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-1	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

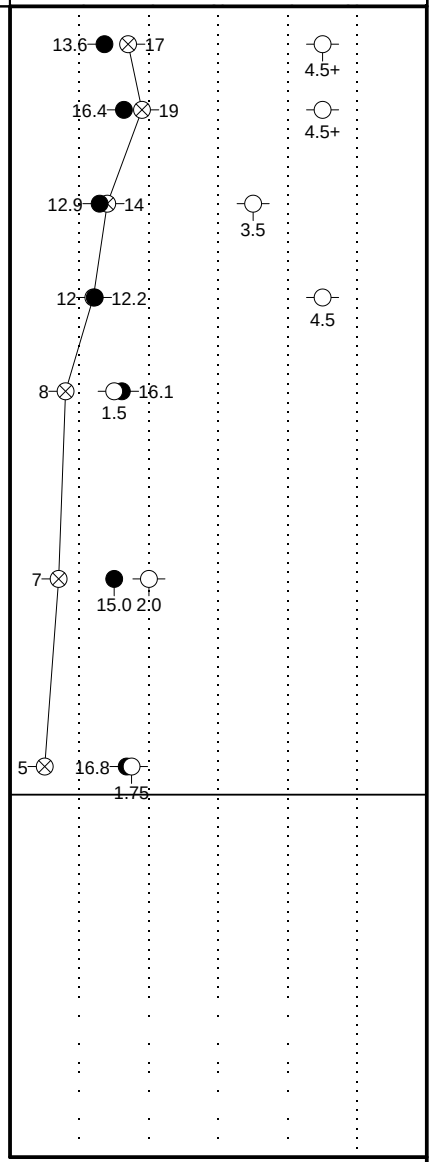
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 719 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	24	24	Topsoil Depth [12"]			5 7 10
					(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown, Dry to Moist, Hard to Very Stiff			13
	S-2	SS	18	18				5 8 11
							715	
5	S-3	SS	18	18				3 6 8
	S-4	SS	18	10	(CL/ML) SILTY CLAY, Trace to Little Sand and Gravel, Gray, Moist, Hard to Stiff			4 6 6
							710	
10	S-5	SS	18	18				3 3 5
							705	
15	S-6	SS	18	18				2 3 4
							700	
20	S-7	SS	18	18				3 3 2
					END OF BORING @ 21'			
25							695	
							690	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL Dry WS ☐ WD ☒	BORING STARTED 08/05/16	CAVE IN DEPTH 16
WL(SHW) WL(ACR) Dry	BORING COMPLETED 08/05/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-2	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT^2

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

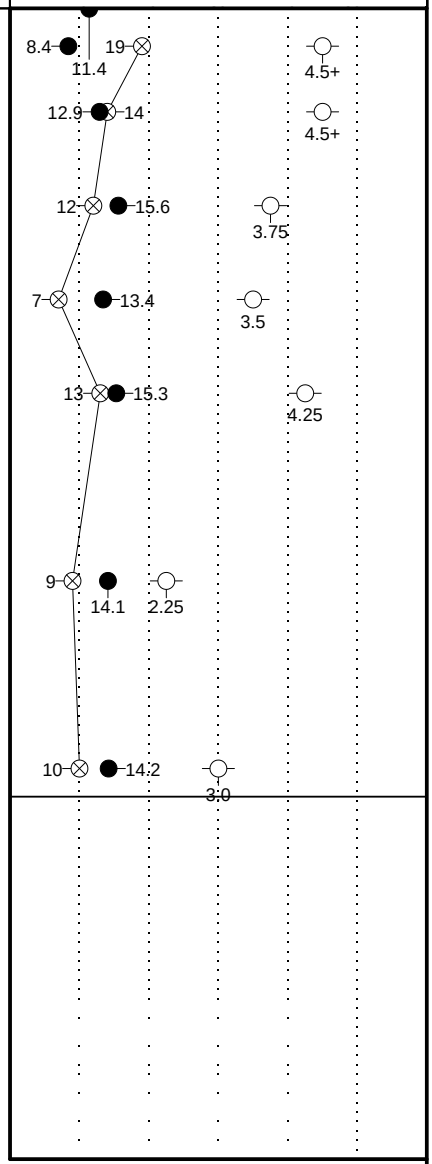
STANDARD PENETRATION BLOWS/FT

DESCRIPTION OF MATERIAL — ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 715.5 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	24	24	Topsoil Depth [12"]		715	3 7 12
					(CL/ML FILL) SILTY CLAY, Trace Sand and Gravel, Brown and Dark Brown, Dry to Moist, Hard			11
	S-2	SS	18	16				3 7 7
5	S-3	SS	18	12	(CL/ML POSSIBLE BURIED TOPSOIL) SILTY CLAY, Trace Sand and Gravel, Trace Organic Matter, Dark Brown, Moist, Very Stiff		710	4 5 7
	S-4	SS	18	15	(ML/CL) CLAYEY SILT, Trace Gravel, Trace Sand, Brown and Gray Mottled, Moist to Wet, Very Stiff			2 3 4
10	S-5	SS	18	16	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Gray, Moist to Wet, Hard to Very Stiff		705	4 5 8
15	S-6	SS	18	18			700	2 3 6
20	S-7	SS	18	18			695	3 4 6
					END OF BORING @ 21'			
25							690	
30							685	



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL Dry WS ☐ WD ☒	BORING STARTED 08/05/16	CAVE IN DEPTH 17
WL(SHW) WL(ACR) Dry	BORING COMPLETED 08/05/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-3	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

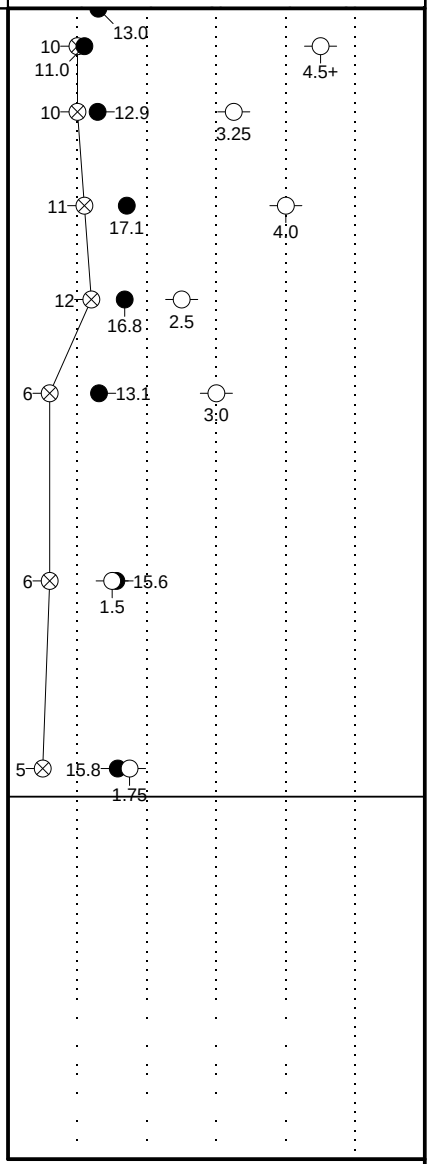
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING ▶ | LOSS OF CIRCULATION

SURFACE ELEVATION 707.5 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	24	19	Topsoil Depth [13"]			3 5 5
					(CL/ML FILL) SILTY CLAY, Trace to Little Sand and Gravel, Brown and Dark Brown, Dry to Moist, Hard			7 4 5
	S-2	SS	18	18	(SM/SC FILL) SILTY/CLAYEY SAND, Trace Gravel, Brown, Moist, Medium Dense		705	5
5	S-3	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown and Gray Mottled, Moist, Hard			3 5 6
	S-4	SS	18	18	(ML) SILT, Trace Clay, Little Sand, Brown and Gray Mottled, Wet, Very Stiff		700	4 7 5
10	S-5	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Gray, Moist to Wet, Very Stiff to Stiff			2 3 3
							695	
15	S-6	SS	18	18				2 2 4
							690	
20	S-7	SS	18	18				2 2 3
					END OF BORING @ 21'		685	
25							680	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL 9.5 WS ☐ WD ☒	BORING STARTED 08/05/16	CAVE IN DEPTH 17.0'
WL(SHW) WL(ACR) 15.0	BORING COMPLETED 08/05/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-4	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

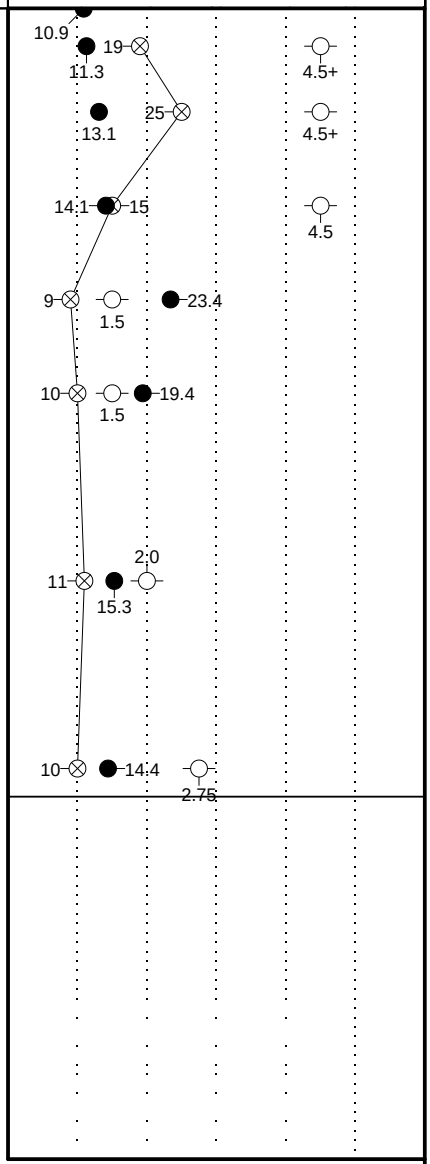
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 716 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0					Topsoil Depth [8"]			4 8 11 13
	S-1	SS	24	19	(CL/ML FILL) SILTY CLAY, Trace Sand, Trace Gravel, Brown, Dry, Hard		715	5 10 15
	S-2	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown and Gray and Light Brown, Dry to Moist, Very Stiff			4 7 8
5	S-3	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown and Slightly Gray Mottled, Moist, Hard		710	3 4 5
	S-4	SS	18	15	(ML/CL) CLAYEY SILT, Trace Sand, Light Brown to Gray at 9.5 feet, Wet, Stiff			3 5 5
10	S-5	SS	18	15			705	3 5 6
15	S-6	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Gray, Moist, Very Stiff		700	2 4 6
20	S-7	SS	18	18				
					END OF BORING @ 21'		695	
25							690	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL Dry WS ☐ WD ☒	BORING STARTED 08/05/16	CAVE IN DEPTH 18'
WL(SHW) WL(ACR) Dry	BORING COMPLETED 08/05/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-5	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

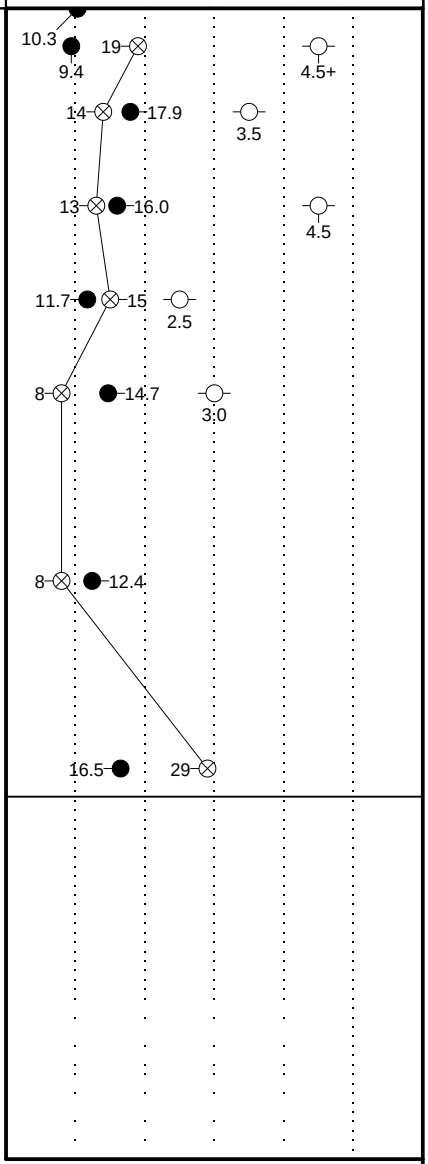
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 715 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	24	24	Topsoil Depth [15"]		715	4 8 11 15
	S-2	SS	18	18	(CL/ML FILL) SILTY CLAY, Little Sand and Gravel, Brown and Light Brown and Dark Brown, Dry, Hard			6 6 8
5	S-3	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown and Yellowish Brown and Gray Mottled, Moist, Very Stiff to Hard		710	3 5 8
	S-4	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Gray, Moist, Very Stiff			5 7 8
10	S-5	SS	18	12			705	4 4 4
15	S-6	SS	18	15			700	5 4 4
20	S-7	SS	18	10	(SM) SILTY SAND, Trace Clay, Gray, Wet, Medium Dense	▽	695	8 14 15
					END OF BORING @ 21'			
25							690	
30							685	



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL 19.5 WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 15.0'
WL(SHW) WL(ACR) Dry	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-6	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION

108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT^2

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

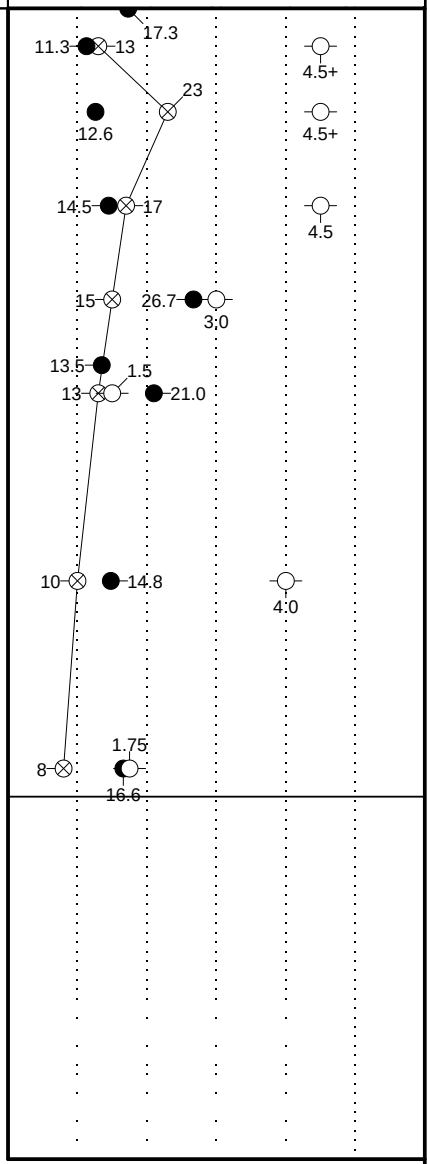
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING ▶ | LOSS OF CIRCULATION

SURFACE ELEVATION 716.5 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	24	24	Topsoil Depth [6"] (CL/ML FILL) SILTY CLAY, Trace Gravel, Trace Sand, Brown, Dry, Hard		715	6 5 8 9
	S-2	SS	18	18				4 10 13
5	S-3	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brown and Gray Mottled, Moist, Hard		710	4 7 10
	S-4	SS	18	16	(ML/CL) CLAYEY SILT with Sand Seams, Brown, Wet, Stiff			5 7 8
10	S-5	SS	18	15	(SM) SILTY SAND, Little Gravel, Brown, Wet, Medium Dense (ML/CL) CLAYEY SILT, Trace Sand, Gray, Wet, Stiff		705	4 6 7
15	S-6	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Gray, Moist to Wet, Stiff	▽	700	3 4 6
20	S-7	SS	18	18		▽		2 3 5
					END OF BORING @ 21'		695	
25							690	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

▽ WL 19.5 WS ☐ WD ☒	BORING STARTED 08/05/16	CAVE IN DEPTH @ 19.0'
▽ WL(SHW) ▽ WL(ACR) 16.0	BORING COMPLETED 08/05/16	HAMMER TYPE Auto
▽ WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-7	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

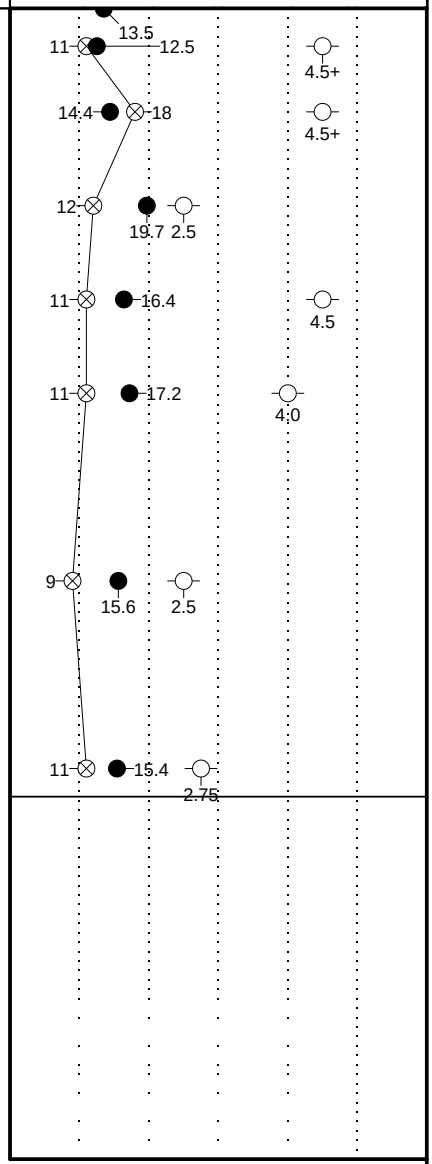
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 710 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0					Topsoil Depth [10"]		710	4 5 6
	S-1	SS	24	15	(CL/ML FILL) SILTY CLAY, Little Sand, Trace Gravel, Brown and Dark Brown, Dry, Hard			5 6
	S-2	SS	18	16	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brown and Gray Mottled, Moist, Hard			8 10
5	S-3	SS	18	18	(ML/CL) CLAYEY SILT, Little Sand, Brown and Gray Mottled, Wet, Very Stiff		705	3 6 6
	S-4	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Brown and Gray Mottled, Moist, Hard			4 6 5
					(SM) SILTY SAND, Brown, Wet, Medium Dense			
10	S-5	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Gray, Moist to Wet, Very Stiff		700	3 4 7
15	S-6	SS	18	18			695	3 3 6
20	S-7	SS	18	18			690	3 5 6
					END OF BORING @ 21'			
25							685	
30							680	



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL Dry WS ☐ WD ☒	BORING STARTED 08/05/16	CAVE IN DEPTH 19
WL(SHW) WL(ACR) Dry	BORING COMPLETED 08/05/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-8	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

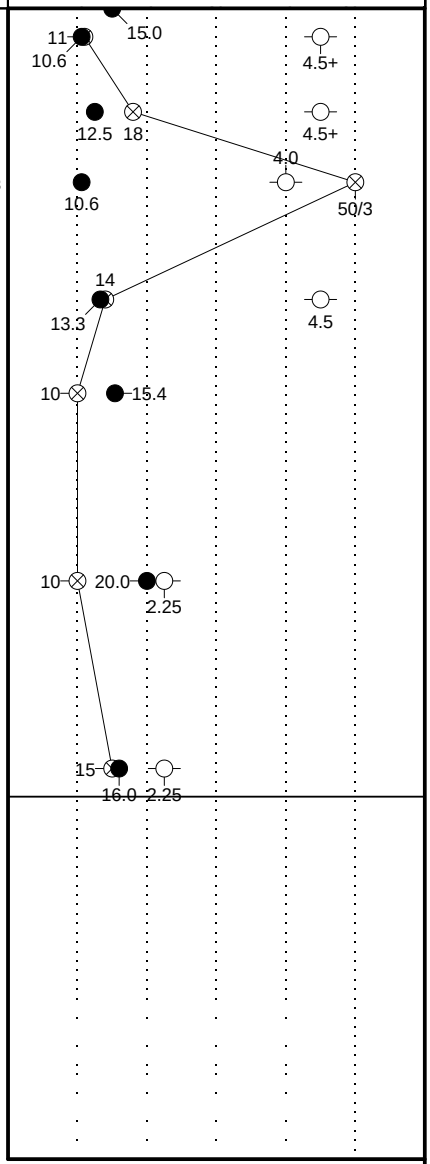
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING ▶ | LOSS OF CIRCULATION

SURFACE ELEVATION 716 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	18	18	Topsoil Depth [12"]		715	3 5 6
					(CL/ML) SILTY CLAY, Trace to Little Sand and Gravel, Light Brown, Dry to Moist, Hard			
	S-2	SS	18	18	(ML/CL) CLAYEY SILT, Trace Gravel, Trace Sand, Brown and Grayish Brown, Moist, Hard			5 9 9
5	S-3	SS	3	3			710	50/3
	S-4	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brownish Gray, Moist, Hard			5 9 5
10	S-5	SS	18	10	(SC) CLAYEY SAND, Some Silt, Brown, Wet, Medium Dense	▽	705	3 5 5
15	S-6	SS	18	18	(CL/ML) SILTY CLAY to CLAYEY SILT, Trace Sand, Gray, Wet, Very Stiff (Varved)		700	6 5 5
20	S-7	SS	18	16			695	6 7 8
					END OF BORING @ 21'			
25							690	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL 9.5 WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 16.0'
WL(SHW) WL(ACR) 14.0	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-9	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT^2

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

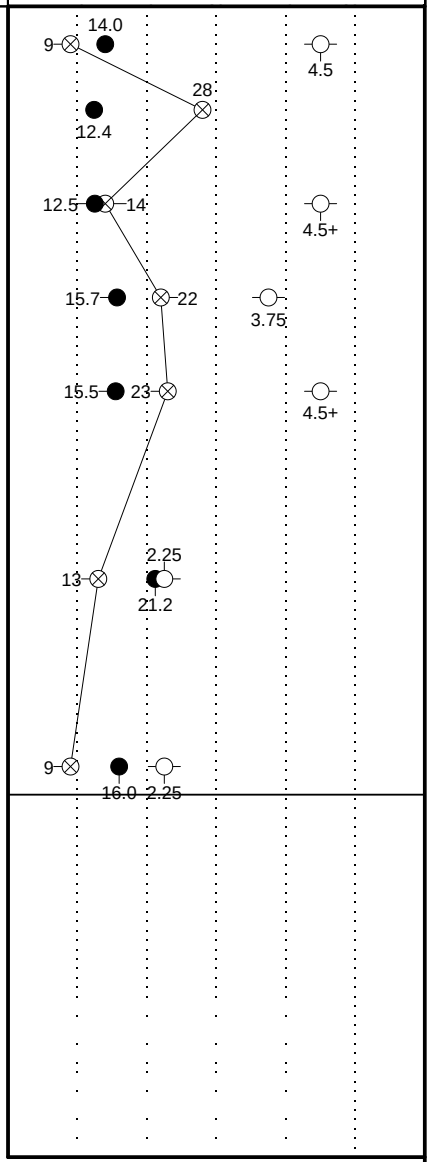
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING ▰ LOSS OF CIRCULATION (100%)

SURFACE ELEVATION 718 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0					Topsoil Depth [8"]			2
	S-1	SS	24	24	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown, Possible Cobbles between 2.5 to 4.5 +/- feet, Dry, Hard			3 6 6
	S-2	SS	18	18			715	21 19 9
5	S-3	SS	18	13	(CL/ML) SILTY CLAY, Little Sand and Gravel, Occasional Sand Seams, Brown and Light Brown, Moist, Hard			6 6 8
	S-4	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown, Moist, Very Stiff to Hard		710	6 9 13
10	S-5	SS	18	18				7 11 12
							705	
15	S-6	SS	18	18	(ML/CL) CLAYEY SILT, Trace Sand, Gray, Wet, Very Stiff	▽		4 5 8
							700	
20	S-7	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Gray, Moist to Very Moist, Very Stiff			3 4 5
					END OF BORING @ 21'		695	
25								
							690	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

▽ WL 14.5 WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 18.0'
▽ WL(SHW) ▼ WL(ACR) 15.0	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
▽ WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-10	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

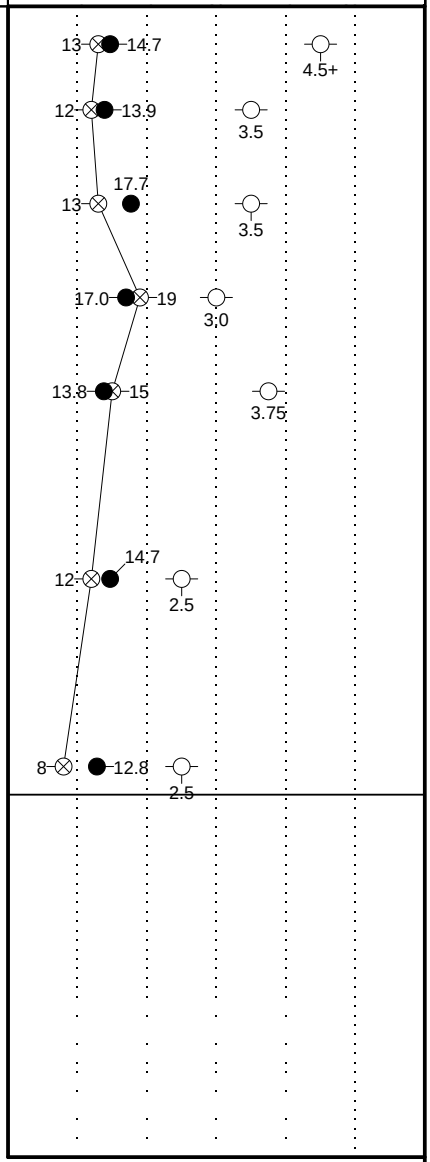
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 721 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0					Topsoil Depth [8"]			4
	S-1	SS	24	22	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brown, Dry to Moist, Hard to Very Stiff		720	6 7 10
	S-2	SS	18	18				4 5 7
5	S-3	SS	18	18			715	5 5 8
	S-4	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brown and Slightly Mottled Gray , Moist, Very Stiff			7 9 10
10	S-5	SS	18	15	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Gray, Moist to Wet, Very Stiff		710	5 7 8
15	S-6	SS	18	18			705	4 5 7
20	S-7	SS	18	18				3 4 4
					END OF BORING @ 21'		700	
25							695	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL Dry WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 18'
WL(SHW) WL(ACR) Dry	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-11	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

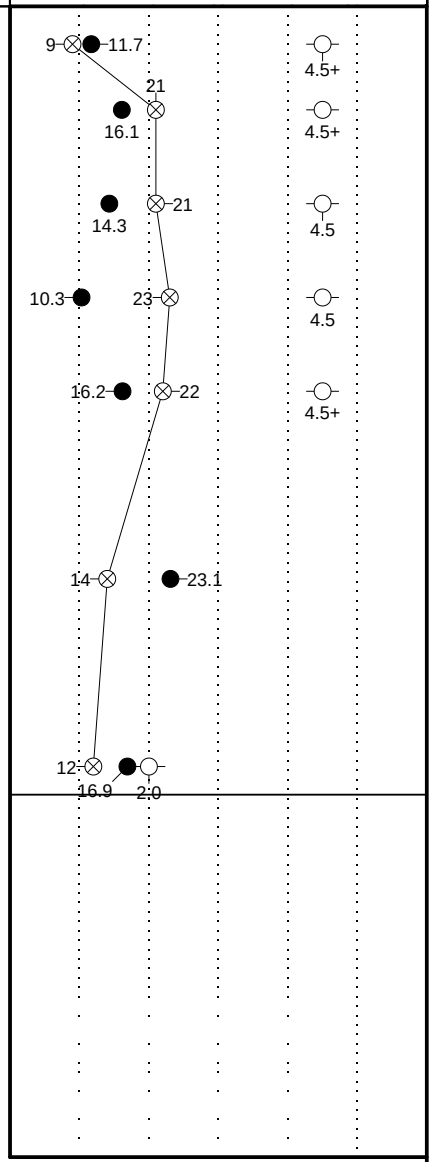
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 725.5 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0					Topsoil Depth [8"]		725	4
	S-1	SS	24	22	(CL/ML) SILTY CLAY, Trace Sand, Brown, Dry to Moist, Hard			4 5 6
	S-2	SS	18	18				6 8 9 12
5	S-3	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brown and Light Brown and Gray Mottled, Moist, Hard		720	7 10 11
	S-4	SS	18	18				8 12 11
10	S-5	SS	18	18	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Gray, Moist to Wet, Very Stiff		715	8 10 12
15	S-6	SS	18	16	(SM/ML) SILTY SAND/SANDY SILT, Gray, Wet, Medium Dense	▼	710	5 6 8
20	S-7	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Gray, Moist, Very Stiff		705	4 5 7
					END OF BORING @ 21'			
25							700	
30							695	



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL 14.5 WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 16.0'
WL(SHW) WL(ACR) 15.0	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-12	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

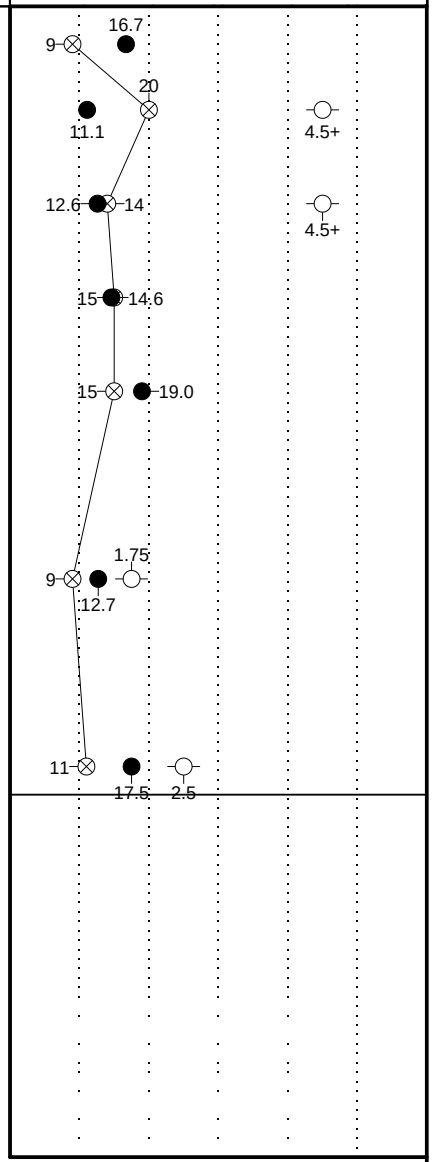
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

SURFACE ELEVATION 722.5 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0					Topsoil Depth [8"]			3
	S-1	SS	24	21	(CL/ML) SILTY CLAY, Trace Gravel, Trace Sand, Brown, Moist, Very Stiff			4 5
	S-2	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand and Gravel, Orange and Orangish Brown and Gray Mottled, Dry to Moist, Hard		720	5 8 8 9 11
5	S-3	SS	18	18	(CL/ML) SILTY CLAY, Little to Some Sand and Gravel, Brown, Moist, Hard			5 6 8
	S-4	SS	18	15	(SC) CLAYEY SAND, Trace Gravel, Grayish Brown, Very Moist to Wet, Medium Dense		715	6 6 9
10	S-5	SS	18	18	(ML/CL) CLAYEY SILT, Trace Sand, Gray, Wet, Stiff	▽		5 7 8
						▼	710	
					(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Gray, Very Moist, Stiff to Very Stiff			
15	S-6	SS	18	18			705	5 4 5
20	S-7	SS	18	18				4 4 7
					END OF BORING @ 21'		700	
25							695	
30								



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL 9.5 WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 12.0'
WL(SHW) WL(ACR) 11.0	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA

CLIENT	JOB #	BORING #	SHEET
Zilber Property Group	42:1291	B-13	1 OF 1

PROJECT NAME	ARCHITECT-ENGINEER
ZPG Nitto Denko Remnant Property	Pinnacle Engineering Group

SITE LOCATION: 108th Street & Highway 31, Pleasant Prairie, Kenosha County, WI

NORTHING | EASTING | STATION

CALIBRATED PENETROMETER TONS/FT²

ROCK QUALITY DESIGNATION & RECOVERY
RQD% – — – REC% ———

PLASTIC LIMIT% | WATER CONTENT% | LIQUID LIMIT%

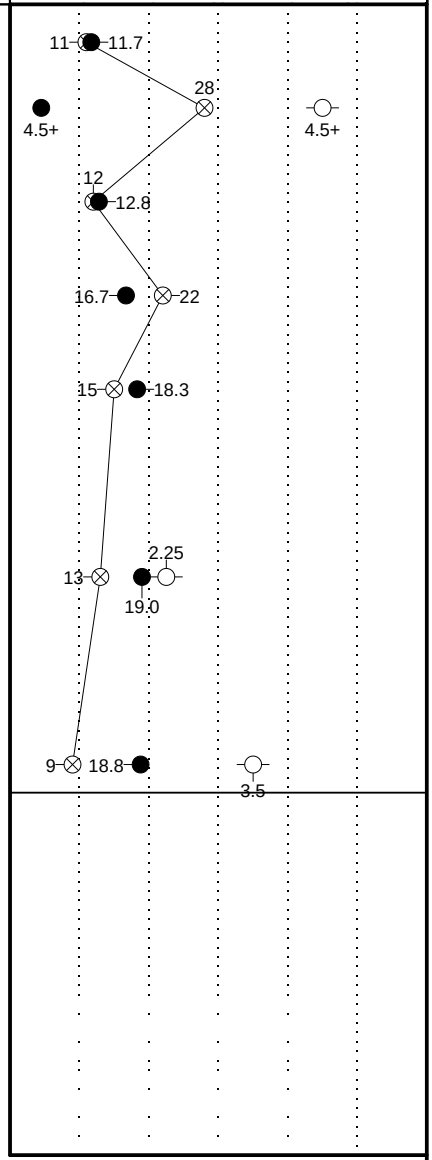
STANDARD PENETRATION BLOWS/FT

ENGLISH UNITS

BOTTOM OF CASING | LOSS OF CIRCULATION

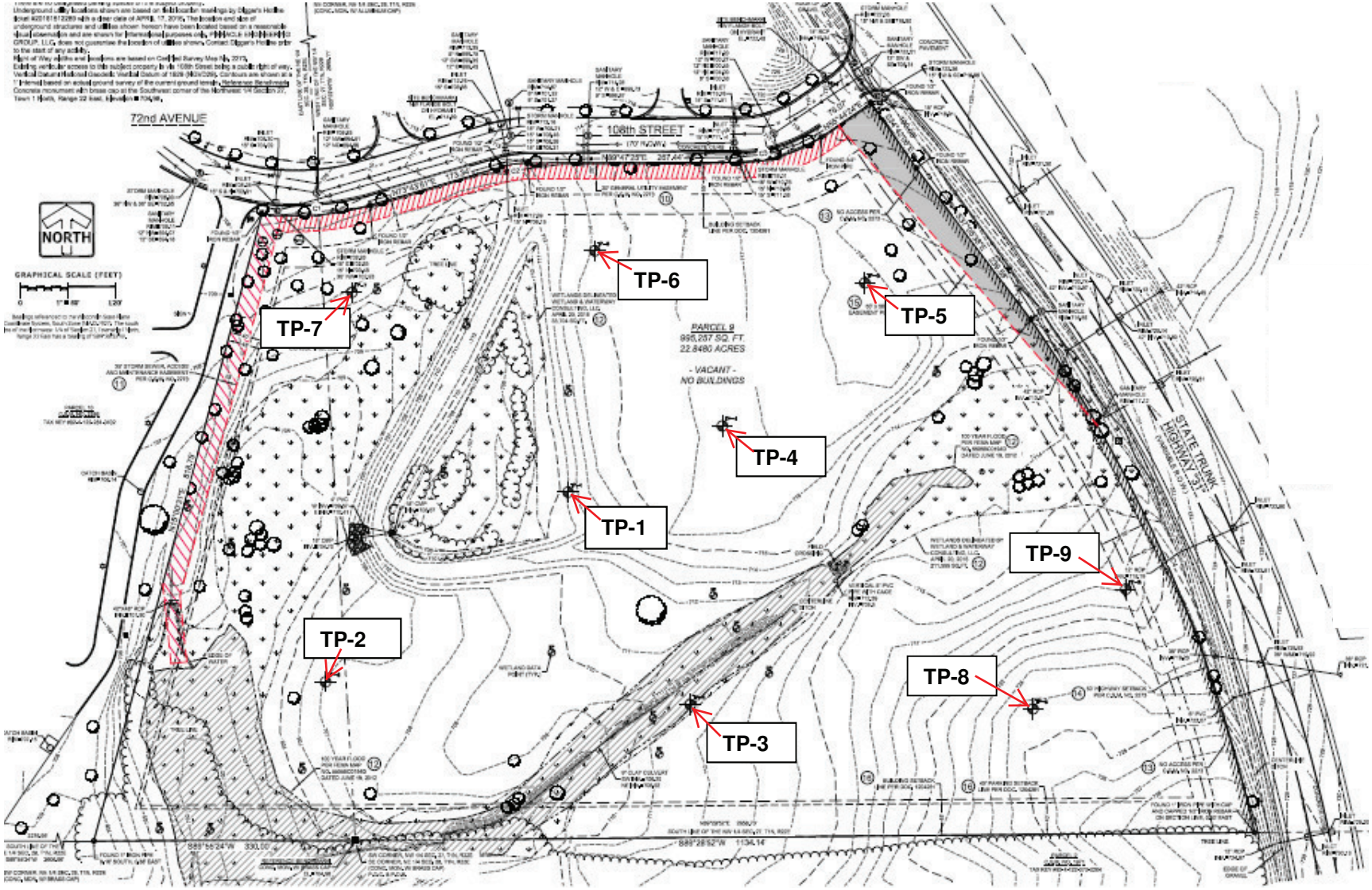
SURFACE ELEVATION 715 +/-

DEPTH (FT)	SAMPLE NO.	SAMPLE TYPE	SAMPLE DIST. (IN)	RECOVERY (IN)	DESCRIPTION OF MATERIAL	WATER LEVELS	ELEVATION (FT)	BLOWS/6"
0	S-1	SS	24	5	Topsoil Depth [24"]		715	2 5 6 7
	S-2	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Brown and Orangish Brown and Gray Mottled, Dry to Moist, Hard			10 15 13
5	S-3	SS	18	15	(SM) SILTY SAND, Trace Clay, Trace Gravel, Brown, Very Moist to Wet, Medium Dense		710	4 6 6
	S-4	SS	18	16				8 10 12
10	S-5	SS	18	18	(ML/CL) CLAYEY SILT, Trace to Little Sand, Gray, Wet, Medium Dense	▽	705	7 7 8
15	S-6	SS	18	18	(CL/ML) SILTY CLAY, Trace Sand, Trace Gravel, Gray, Moist to Wet, Very Stiff	▼	700	5 6 7
20	S-7	SS	18	18			695	5 4 5
					END OF BORING @ 21'			
25							690	
30							685	



THE STRATIFICATION LINES REPRESENT THE APPROXIMATE BOUNDARY LINES BETWEEN SOIL TYPES. IN-SITU THE TRANSITION MAY BE GRADUAL.

WL 9.5 WS ☐ WD ☒	BORING STARTED 08/08/16	CAVE IN DEPTH @ 17.0'
WL(SHW) WL(ACR) 15.0	BORING COMPLETED 08/08/16	HAMMER TYPE Auto
WL	RIG Truck FOREMAN Kyle	DRILLING METHOD HSA



TEST PIT LOCATION PLAN.
Proposed Business Park
ZPG Nitto Denko Remnant Property
108th Street and Highway 31
Pleasant Prairie, Wisconsin
ECS Project No. 42:1291

Notes:

1) TP-1 = Test Pit Location.
2) Not to Scale
3) Plan developed from Plat of Survey (Soil Borings Revision dated 08-18-16) prepared by Pinnacle Engineering Group

TEST PIT SUMMARY
Proposed Business Park
ZPG Nitto Remnant Property
SWC 108th Street and Highway 31
Pleasant Prairie, Wisconsin
ECS Project No. 42:1291
Page 1

Test Pit Number	Depth Below Existing Site Grade (Feet)	Soil Conditions	Remarks
TP-1 (EL. +711)[1]	0 to 0.8±	Topsoil (9½± inches)	Groundwater was observed during excavation at 9± feet. Water level at 8½± feet at 2½ ± hours.
	0.8 to 2.2±	Dark Brown Silty CLAY, trace Sand and Gravel, Dry to Moist (**FILL**) (Qp = 4½+ tsf, w = 17.7 percent)	
	2.2 to 6±	Brown and Gray mottled Silty CLAY, trace Sand and Gravel, Moist (Qp = 4½+ tsf, w = 12.4 percent)	
	6 to 7½±	Brown Clayey SILT, little to some fine Sand , Very Moist (Qp = 2¼ tsf, w = 18.0 percent)	
	7½ to 9±	Brown Fine Sandy CLAY, Wet (w = 18.0 percent)	
TP-2 (EL. +707)	0 to 0.9±	Topsoil (10± inches)	No perched water or groundwater was observed at completion or at 2± hours after completion.
	0.9 to 1½ ±	Brown Silty CLAY, trace Sand and Gravel, Dry (w = 6.7 percent)	
	1½ to 5±	Brown and Dark Brown Silty CLAY, trace Sand and Gravel, Moist (Qp = 4½+ tsf, w = 21.2 percent)	
	5 to 6±	Brown Silty CLAY, trace to little Sand and Gravel, Moist	
	6 to 8½±	Grayish Brown Silty CLAY, trace to little Sand and Gravel, Moist (Qp = 4½+ tsf, w = 14.7 and 16.0 percent)	
TP-3 (EL. +711)	0 to 3±	Topsoil (36± inches)	No perched water or free groundwater was observed during excavation. Very Moist at 6± feet. Test pit backfilled immediately upon completion of excavation.
	3 to 6	Brownish Gray Silty CLAY, trace to little Sand and Gravel, Very Moist (Qp = 3¾ tsf, w = 17.0 percent)	
TP-4 (EL. +716.5)	0 to ¾±	Topsoil (9± inches)	No perched water or groundwater was observed at completion or at 1¾± hours after completion
	¾ to 3±	Brown and Dark Brown Silty CLAY, trace Sand and Gravel, Dry to Moist (**FILL**) (Qp = 4½+ tsf, w = 9.5 to 10.0 percent)	
	3 to 4±	Brown and Gray mottled Silty CLAY, trace Sand and Gravel, Moist (Qp = 4½+ tsf, w = 15.7 percent)	

Note 1: Elevations are interpreted from Plat of Survey (dated 05-02-2016, Revision 1 Soil Borings dated 08-09-16) prepared by Pinnacle Engineering Group.
Qp = Unconfined compressive strength using hand penetrometer (tsf = tons per square foot)
w = moisture content

TEST PIT SUMMARY
Proposed Business Park
ZPG Nitto Remnant Property
SWC 108th Street and Highway 31
Pleasant Prairie, Wisconsin
ECS Project No. 42:1291

Test Pit Number	Depth Below Existing Site Grade (Feet)	Soil Conditions	Remarks
TP-5 (EL. +718.4)	0 to 1±	Topsoil (12± inches)	No perched water or groundwater was observed at completion or at 1± hour after completion.
	1 to 3±	Brown Silty CLAY, trace Sand and Gravel, Dry to Moist (**FILL**) (Qp = 4½+ tsf, w = 11.6 percent))	
	3 to 4±	Brown Silty CLAY, trace Sand and Gravel, Moist (Qp = 4½+ tsf, w = 11.9 percent)	
TP-6 (EL. +714.1)	0 to 1.3±	Topsoil (16± inches)	No perched water or groundwater was observed at completion or at 1± hours after completion.
	1.3 to 3½ ±	Brown and Dark Brown Silty CLAY, trace Sand and Gravel, Moist (**FILL**) (Qp = 4½+ tsf, w = 9.5 percent)	
	3½ to 4±	Brown Silty CLAY, trace Sand and Gravel, Moist (Qp = 4½+ tsf, w = 10.0 percent)	
TP-7 (EL. +708.7)	0 to 0.8±	Topsoil (10± inches)	Groundwater was observed during excavation at 8½± feet. Test pit backfilled immediately upon completion of excavation.
	0.8 to 2.2±	Brown fine to medium Sandy CLAY, Moist (**FILL**) (Qp = 4½+ tsf, w = 17.7 percent)	
	2.2 to 5½±	Brown and Gray mottled Clayey SILT, trace Sand, Very Moist (Qp = 4½+ to 2¾ tsf, w = 19.1 and 17.8 percent)	
	5½ to 8½±	Gray Silty CLAY, trace Sand and Gravel, Very Moist to Wet (Qp = 3½ to 4¼ tsf, w = 11.7 and 11.8 percent)	
TP-8 (EL. +724.2)	0 to ½±	Topsoil (6± inches)	No perched water or groundwater was observed during or at completion of test pit. Test pit backfilled immediately upon completion of excavation.
	½ to 7±	Brown Silty CLAY, trace Sand and Gravel, Dry to Moist (Qp = 4½ tsf, w = 16.3 percent)	
TP-9 (EL. +718)	0 to 1.3±	Topsoil (16± inches)	Very Moist to wet soils observed during excavation at 7½± feet. Test pit backfilled immediately upon completion of excavation.
	1.3 to 5±	Brown Silty CLAY, trace Sand and Gravel, Dry to Moist (Qp = 4½+ tsf, w = 12.8 and 14.8 percent)	
	5 to 7½±	Brown and Gray mottled Silty CLAY, little Sand and Gravel, Moist (Qp = 2¼ tsf, w = 18.9 percent)	
	7½ to 8±	Gray Silty CLAY, trace Sand and Gravel, Very Moist to Wet (Qp = 4½ tsf, w = 13.2 percent)	

Wis. Dept. of Safety and Professional Services
Division of Industry Services

SOIL EVALUATION - STORM

in accordance with SPS 382.365 and 385, Wis. Adm. Code

Attach complete site plan on paper not less than 8½ x 11 inches in size. Plan must include, but not limited to: vertical and horizontal reference point (BM), direction and percent slope, scale or dimensions, north arrow, and BM referenced to nearest road.

Please print all information.

Personal information you provide may be used for secondary purposes (Privacy Law, s, 15.04 (1) (m)).

County	
Parcel I.D.	
Reviewed by	Date

Property Owner **Zilber Property Group**			Property Location **SWC 108th Street and Highway 31** Govt. Lot : PTs of **SW1/4 of the NW ¼ of S. 27** and **SE1/4 of the NE1/4** of S **28** in T **1** N R **22 E**
Property Owner's Mailing Address **9560 58th Street, Suite 350**			Lot # / Block # / Subd. Name or CSM# **Parcel 9 of CSM No. 2273**
City **Kenosha**	State **WI**	Zip Code **53144**	Phone Number (630) 318-6042
☐ City ☑ Village ☐ Town **Pleasant Prairie**			Nearest Road **Highway 31**

Drainage area _____ sq. ft. acres

Optional:

Test Site Suitable for (check all that apply)

Irrigation ☐ Bioretention trench Trench(es)

Rain garden Grassed swale Reuse

Infiltration trench SDS (>15' wide) Other _____

Hydraulic Application Test Method:

☑ Morphological Evaluation

☐ Double-Ring Infiltrometer

Other (specify) _____

TP-2 Obs. #

☐ Boring
☑ Pit Ground surface elev. 707± ft. Depth to limiting factor 0 in.

Horizon	Depth in.	Dominant Color Munsell	Redox Description Qu. Sz. Cont. Color	Texture	Structure Gr. Sz. Sh.	Consistence	Boundary	% Rock Frag.	Hydraulic App. Rate Inches/Hr.
A	0 - 12	10 YR 3/2		SiCL	2, abk, vf	dvh	w	< 5	0.04
C1	12 - 18	10 YR 4/4		SiCL	3, abk, f-m	dvh	c	< 5	0.04
C2	18 - 60	10 YR 5/6	10 YR 7/2 f, 1, f	SiC	3, abk f,	dvh	c	< 5	0.07
C3	60 - 72	10 YR 5/3		SiCL	2, abk, f	mfi	c	< 5	0.04
C4	72- 114	10 YR 5/2		SiCL	2, qbk, f	mvfi		< 5	0.04

No free water encountered during exploration. Seasonal high groundwater estimated at 1½± feet (EL. 705.5) based on soil coloration.

TP-3 Obs. #

☐ Boring
☑ Pit Ground surface elev. 711± ft. Depth to limiting factor 0 in.

Horizon	Depth in.	Dominant Color Munsell	Redox Description Qu. Sz. Cont. Color	Texture	Structure Gr. Sz. Sh.	Consistence	Boundary	% Rock Frag.	Hydraulic App. Rate Inches/Hr.
A	0 - 36	10 YR 2/1		SiCL	0, sg, f	mfr	w	<5	0.04
C1	36 - 72	10 YR 5/4		SiCL	2, abk, f-m	mfi		<5	0.04

No free water encountered during exploration but soils from 3 to 6 feet were very moist. Seasonal high groundwater estimated at 3± feet (EL. 708) based on soil coloration.

CST/PSS Name (Please Print) Paul J. Giese Signature Paul J. Giese CST/PSS Number 41442

Address 3695 N. 126th St, Unit C, Brookfield, WI 53005 Date Evaluation Conducted 08/04/2016 Telephone Number (262) 286-7557

TP-7 Obs. #

□ Boring
☑ Pit Ground surface elev. 708.7± ft. Depth to limiting factor 0 in.

Horizon	Depth in.	Dominant Color Munsell	Redox Description Qu. Sz. Cont. Color	Texture	Structure Gr. Sz. Sh.	Consistence	Boundary	% Rock Frag.	Hydraulic App. Rate Inches/Hr.
FILL	0 - 10	10 YR 4/3		SiCL	2, abk, f	dsh	w	< 5	0.04
FILL	10 - 26	10 YR 4/3		SCL	1, sbk, f	mfr	c	< 5	0.11
C1	26- 66	10 YR 6/4	10 YR 6/2 c, 2, d	CL	2, abk f,	mfr	c	< 5	0.03
C2	66 - 102	10 YR 5/2		SiC	3, abk, f	mvfi	c	< 5	0.07

Free water encountered at 8 ½ feet during exploration. Seasonal high groundwater estimated at 2¼± feet (EL. 706.5±) based on soil coloration.

TP-9 Obs. #

□ Boring
☑ Pit Ground surface elev. 718± ft. Depth to limiting factor 0 in.

Horizon	Depth in.	Dominant Color Munsell	Redox Description Qu. Sz. Cont. Color	Texture	Structure Gr. Sz. Sh.	Consistence	Boundary	% Rock Frag.	Hydraulic App. Rate Inches/Hr.
A	0 - 16	10 YR 3/2		SiCL	3, abk, f	dvh	w	< 5	0.04
C1	16 - 60	10 YR 4/3		SiCL	3, abk, f	dvh	c	< 5	0.04
C2	60 - 90	10 YR 5/3	10 YR 5/2 c, 2, d	SiC	3, m, f-c	mvfi	c	< 5	0.07
C3	90 - 96	10 YR 5/1		SiC	3, m, f	mefi	c	< 5	0.07

No free water encountered during exploration. Seasonal high groundwater estimated at 5± feet (EL. 713) based on soil coloration.

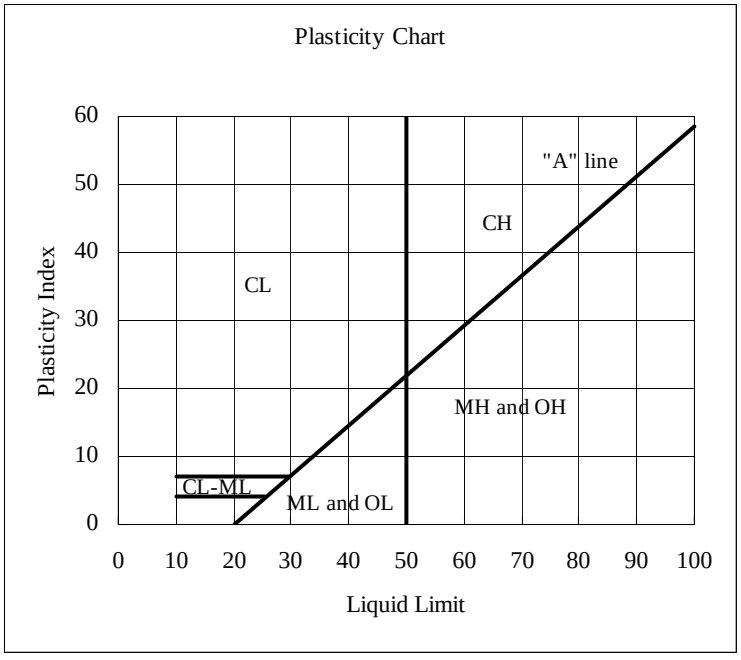
Test Results and/or Summary Comments

The encountered soils consist of predominantly silty clay loam, silty clay and clay loam soils that are considered to have very low infiltration rates of 0.04 inches per hour. 0.07 inches per hour and 0.03 inches per hour , respectively. Based on the encountered silty clay loam, silty clay and clay loam soils, the site is considered to be exempt from infiltration per NR 151.12(5) (c)6.a.

The Dept. of Safety and Professional is an equal opportunity service provider and employer. If you need assistance to access services or need material in an alternate format, contact the department at 608-266-3151 or TTY through Relay

UNIFIED SOIL CLASSIFICATION SYSTEM (ASTM D 2487)

Major Divisions			Group Symbols		Typical Names	Laboratory Classification Criteria		
Coarse-grained soils (More than half of material is larger than No. 200 Sieve size)	Gravels (More than half of coarse fraction is larger than No. 4 sieve size)	Clean gravels (Little or no fines)	GW		Well-graded gravels, gravel-sand mixtures, little or no fines	Determine percentages of sand and gravel from grain-size curve. Depending on percentage of fines (fraction smaller than No. 200 sieve size), coarse-grained soils are classified as follows: Less than 5 percent GW, GP, SW, SP; More than 12 percent GM, GC, SM, SC; 5 to 12 percent Borderline cases requiring dual symbols [b]	$C_u = D_{60}/D_{10}$ greater than 4 $C_c = (D_{30})^2/(D_{10}xD_{60})$ between 1 and 3	
			GP		Poorly graded gravels, gravel-sand mixtures, little or no fines		Not meeting all gradation requirements for GW	
		Gravels with fines (Appreciable amount of fines)	GM [a]	d	Silty gravels, gravel-sand mixtures		Atterberg limits below "A" line or P.I. less than 4	Above "A" line with P.I. between 4 and 7 are borderline cases requiring use of dual symbols
				u				
			GC		Clayey gravels, gravel-sand-clay mixtures		Atterberg limits below "A" line or P.I. less than 7	
	Sands (More than half of coarse fraction is smaller than No. 4 sieve size)	Clean sands (Little or no fines)	SW		Well-graded sands, gravelly sands, little or no fines		$C_u = D_{60}/D_{10}$ greater than 6 $C_c = (D_{30})^2/(D_{10}xD_{60})$ between 1 and 3	
			SP		Poorly graded sands, gravelly sands, little or no fines		Not meeting all gradation requirements for SW	
		Sands with fines (Appreciable amount of fines)	SM [a]	d	Silty sands, sand-silt mixtures		Atterberg limits above "A" line or P.I. less than 4	Limits plotting in CL-ML zone with P.I. between 4 and 7 are borderline cases requiring use of dual symbols
				u				
			SC		Clayey sands, sand-clay mixtures		Atterberg limits above "A" line with P.I. greater than 7	
Fine-grained soils (More than half material is smaller than No. 200 Sieve)	Silts and clays (Liquid limit less than 50)		ML		Inorganic silts and very fine sands, rock flour, silty or clayey fine sands, or clayey silts with slight plasticity			
			CL		Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays			
			OL		Organic silts and organic silty clays of low plasticity			
	Silts and clays (Liquid limit greater than 50)		MH		Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts			
			CH		Inorganic clays of high plasticity, fat clays			
			OH		Organic clays of medium to high plasticity, organic silts			
	Highly Organic soils		Pt		Peat and other highly organic soils			



[a] Division of GM and SM groups into subdivisions of d and u are for roads and airfields only. Subdivision is based on Atterberg limits; suffix d used when L.L. is 28 or less and the P.I. is 6 or less; the suffix u used when L.L. is greater than 28.

[b] Borderline classifications, used for soils possessing characteristics of two groups, are designated by combinations of group symbols. For example: GW-GC,well-graded gravel-sand mixture with clay binder. (From Table 2.16 - Winterkorn and Fang, 1975)

REFERENCE NOTES FOR BORING LOGS

MATERIALS



ASPHALT



CONCRETE



SUBBASE STONE / GRAVEL



TOPSOIL



FILL Man-placed or disturbed soils



GW **WELL-GRADED GRAVEL**
gravel-sand mixtures, little or no fines



GP **POORLY-GRADED GRAVEL**
gravel-sand mixtures, little or no fines



GM **SILTY GRAVEL**
gravel-sand-silt mixtures



GC **CLAYEY GRAVEL**
gravel-sand-clay mixtures



SW **WELL-GRADED SAND**
gravelly sand, little or no fines



SP **POORLY-GRADED SAND**
gravelly sand, little or no fines



SM **SILTY SAND**
sand-silt mixtures



SC **CLAYEY SAND**
sand-clay mixtures



ML **SILT**
non-plastic to medium plasticity



MH **ELASTIC SILT**
high plasticity



CL **LEAN CLAY**
low to medium plasticity



CH **FAT CLAY**
high plasticity



OL **ORGANIC SILT or CLAY**
non-plastic to low plasticity



OH **ORGANIC SILT or CLAY**
high plasticity



PT **PEAT**
highly organic soils



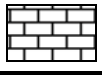
WEATHERED ROCK



IGNEOUS ROCK



METAMORPHIC ROCK



SEDIMENTARY ROCK

DRILLING SAMPLING SYMBOLS & ABBREVIATIONS

SS	Split Spoon Sampler	PM	Pressuremeter Test
ST	Shelby Tube Sampler	RD	Rock Bit Drilling
WS	Wash Sample	RC	Rock Core, NX, BX, AX
BS	Bulk Sample of Cuttings	REC	Rock Sample Recovery %
PA	Power Auger (no sample)	RQD	Rock Quality Designation
HSA	Hollow Stem Auger		

PARTICLE SIZE IDENTIFICATION

DESIGNATION		PARTICLE SIZES
Boulders		12-inches (300-mm) or larger
Cobbles		3-inches to 12- inches (75-mm to 300-mm)
Gravel:	Coarse	¾-inch to 3-inches (19-mm to 75-mm)
	Fine	4.75-mm to 19-mm (No. 4 sieve to ¾-inch)
Sand:	Coarse	2.00-mm to 4.75-mm (No. 10 to No. 4 sieve)
	Medium	0.425-mm to 2.00-mm (No. 40 to No. 10 sieve)
	Fine	0.074-mm to 0.425-mm (No. 200 to No. 40 sieve)
Silt & Clay ("Fines")		<0.074-mm (smaller than a No. 200 sieve)

WATER LEVELS[1]

	WL	Water Level (WS)(WD) (WS) While Sampling (WD) While Drilling
	BCR	Before Casing Removal
	ACR	After Casing Removal
	WL	Water Level as stated
	DCI	Dry Cave-In
	WCI	Wet Cave-In

RELATIVE PROPORTIONS

Trace	<5%
Little	5% - <15%
With	15% - <30%
Adjective *(ex: "Silty")*	30% - <50%

COHESIVE SILTS & CLAYS

UNCONFINED COMP. STRENGTH, Q_P[2] (TSF)	SPT[3] (BPF)	CONSISTENCY (COHESIVE ONLY)
<0.25	≤2	Very Soft
0.25 - 0.49	3 - 4	Soft
0.50 - 0.99	5 - 8	Medium Stiff
1.00 - 1.99	9 - 15	Stiff
2.00 - 3.99	16 - 30	Very Stiff
4.00 - 8.00	31 - 50	Hard
>8.00	>50	Very Hard

GRAVELS, SANDS & NON-COHESIVE SILTS

SPT[3] (BPF)	DENSITY
≤4	Very Loose
5 - 10	Loose
11 - 30	Medium Dense
31 - 50	Dense
51 - 99	Very Dense
≥100	Partially Weathered Rock to Intact Rock

[1]The water levels are those levels actually measured in the borehole at the times indicated by the symbol. The measurements are relatively reliable when augering, without adding fluids, in granular soils. In clay and cohesive silts, the determination of water levels may require several days for the water level to stabilize. In such cases, additional methods of measurement are generally taken.

[2]Typically estimated via pocket penetrometer or Torvane shear test and expressed in tons per square foot (tsf).

[3]Standard Penetration Test (SPT) refers to the number of hammer blows (blow count) of a 140 lb. hammer falling 30 inches on a 2-inch OD split-spoon sampler required to drive the sampler 12 inches (ASTM D 1586). "N-value" is another term for "blow count" and is expressed in blows per foot (bpf).

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